

VECTOR ALGEBRA

WORKEDOUT EXAMPLES

W.E-1: Let PN be the Perpendicular from the point $P(1, 2, 3)$ to xy-plane. If OP makes an angle α with positive direction of the Z-axis and ON makes an angle β with the positive direction of x-axis, where O is the Origin (α and β are acute angles) then

A. $\sin \alpha \sin \beta = \frac{2}{\sqrt{14}}$ B. $\cos \alpha \cos \beta = \frac{1}{\sqrt{14}}$

C. $\tan \alpha = \frac{\sqrt{5}}{3}$ D. $\tan \beta = 2$

Sol: (A, C, D)

Let $\overrightarrow{OP} = \vec{r}$ then

$r \sin \alpha \cos \beta = 1 \dots (1)$

$r \sin \alpha \sin \beta = 2 \dots (2)$

$r \cos \alpha = 3 \dots (3)$

On squaring and adding, we get

$r^2 = 14 \Rightarrow r = \pm \sqrt{14}$

Using (1), (2), (3) and (4) we get required result.

W.E-2: In a four-Dimensional space where unit vector along axes are $\vec{i}, \vec{j}, \vec{k}$ and $\vec{l}$ and $\vec{a}_1, \vec{a}_2, \vec{a}_3, \vec{a}_4$ are four non-zero vectors such that no vector can be expressed as linear combination of others and

$(\lambda - 1)(\vec{a}_1 - \vec{a}_2) + \mu(\vec{a}_2 + \vec{a}_3) + r(\vec{a}_3 + \vec{a}_4 - 2\vec{a}_2) + \vec{a}_3 + \delta \vec{a}_4 = \vec{0}$

Then

A. $\lambda = 1$ B. $\mu = -2/3$

C. $r = 2/3$ D. $\delta = 1/3$

Sol: (A, B)

$(\lambda - 1)(\vec{a}_1 - \vec{a}_2) + \mu(\vec{a}_2 + \vec{a}_3) + r(\vec{a}_3 + \vec{a}_4 - 2\vec{a}_2) + \vec{a}_3 + \delta \vec{a}_4 = \vec{0}$

$\Rightarrow (\lambda - 1)\vec{a}_1 + (1 - \lambda + \mu - 2r)\vec{a}_2 + (\mu + r + 1)\vec{a}_3 + (\gamma + \delta)\vec{a}_4 = \vec{0}$

$\vec{a}_1, \vec{a}_2, \vec{a}_3, \vec{a}_4$ are linearly independent

$\therefore \lambda - 1 = 0 \dots (1)$

$1 - \lambda + \mu - 2r = 0 \dots (2)$

$\mu + r + 1 = 0 \dots (3)$

$\gamma + \delta = 0 \dots (4)$

By solving (1), (2), (3) and (4), we get

$\lambda = 1, \mu = \frac{2}{3}; \gamma = -\frac{1}{3}, \delta = \frac{1}{3}$

W.E-3: If $\vec{a}$ and $\vec{b}$ are unit vectors then greatest and

least values of $|\vec{a} + \vec{b}| + |\vec{a} - \vec{b}|$ are

A. 2, -2 B. $4, 4\sqrt{2}$

C. $2\sqrt{2}, 2$ D. $2\sqrt{2}, -2\sqrt{2}$

Sol: (C)

Given $|\vec{a}| = |\vec{b}| = 1$

Let $(\vec{a}, \vec{b}) = \theta$ then

$|\vec{a} + \vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 + 2\vec{a} \cdot \vec{b}$

$= 2 + 2 \cos \theta$

$= 4 \cos^2 \frac{\theta}{2}$

$\therefore |\vec{a} + \vec{b}| = 2 \cos \frac{\theta}{2}$

Similarly, $|\vec{a} - \vec{b}| = 2 \sin \frac{\theta}{2}$

$\therefore |\vec{a} + \vec{b}| + |\vec{a} - \vec{b}| = 2 \left(\cos \frac{\theta}{2} + \sin \frac{\theta}{2} \right)$

Clearly $\theta \in [0, 180^\circ] \Rightarrow \frac{\theta}{2} \in [0, 90^\circ]$

$\therefore$ greatest value = $2\sqrt{2}$, least value = $2(1) = 2$

W.E-4: Let ABCD be a tetrahedron with AB=41, BC=36, CA=7, DA=18, DB=27, DC=13. If 'd' is the distance between the midpoints of edges AB and CD, then

A. The last digit of d^2 is 7

B. $d^2 = 137$

C. The last digit of d^2 is 6

D. $d^2 = 126$

Sol: (A, B)

Take D as the origin $\overrightarrow{DA} = \vec{a}, \overrightarrow{DB} = \vec{b}, \overrightarrow{DC} = \vec{c}$

$|\vec{a}| = 18, |\vec{b}| = 27, |\vec{c}| = 13, |\vec{a} - \vec{b}| = 41, |\vec{b} - \vec{c}| = 36, |\vec{c} - \vec{a}| = 7$

$d^2 = \left(\frac{\vec{a} + \vec{b}}{2} - \frac{\vec{c}}{2} \right)^2$

$$\Rightarrow 4d^2 = |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2\vec{a} \cdot \vec{b} - 2\vec{a} \cdot \vec{c} - 2\vec{b} \cdot \vec{c}$$

$$|\vec{a} - \vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 - 2\vec{a} \cdot \vec{b}$$

$$|\vec{b} - \vec{c}|^2 = |\vec{b}|^2 + |\vec{c}|^2 - 2\vec{b} \cdot \vec{c}$$

$$|\vec{c} - \vec{a}|^2 = |\vec{c}|^2 + |\vec{a}|^2 - 2\vec{c} \cdot \vec{a}$$

using the above results we get $d^2 = 137$

W.E-5: Let $\hat{a}, \hat{b}, \hat{c}$ be unit vectors such that

$\hat{a} + \hat{b} + \hat{c} = \vec{0}$ and x, y, z be distinct integers then

$|x\hat{a} + y\hat{b} + z\hat{c}|$ cannot be equal to

- A) 1 B) $\sqrt{2}$ C) 2 D) 3

sol: (A, B)

$\vec{a}, \vec{b}, \vec{c}$ form an equilateral triangle and angle

between each pair is $\frac{2\pi}{3}$

$$\therefore |x\vec{a} + y\vec{b} + z\vec{c}|^2 = x^2 + y^2 + z^2 - xy - yz - zx$$

$$= \frac{1}{2} [(x-y)^2 + (y-z)^2 + (z-x)^2]$$

$$\text{Minimum value} = \frac{1}{2} [1+1+4] = 3$$

$$\therefore |x\vec{a} + y\vec{b} + z\vec{c}| \geq \sqrt{3}$$

W.E-6: In an Isosceless triangle ABC, $|\vec{AB}| = |\vec{BC}| = 8$.

A point E divides $\vec{AB}$ internally in the ratio 1 : 3 then the angle between $\vec{CE}$ and $\vec{CA}$ where

$|\vec{CA}| = 12$ is

A. $\cos^{-1}\left(\frac{\sqrt{7}}{8}\right)$ B. $\cos^{-1}\left(\frac{3\sqrt{7}}{8}\right)$

C. $\cos^{-1}\left(\frac{\sqrt{7}}{4}\right)$ D. $\cos^{-1}\left(\frac{3\sqrt{7}}{4}\right)$

Sol: (B)

Let 'A' be the fixed point and $\vec{AB} = \vec{b}$, $\vec{AC} = \vec{c}$
Then

$$|\vec{b}| = 8, |\vec{c}| = 8, |\vec{b} - \vec{c}| = 12$$

$$\text{Now, } |\vec{b} - \vec{c}|^2 = 144$$

$$\Rightarrow |\vec{b}|^2 + |\vec{c}|^2 - 2\vec{b} \cdot \vec{c} = 144$$

$$\Rightarrow \vec{b} \cdot \vec{c} = 72$$

$$\text{Since } AE : EB = 1 : 3 \Rightarrow \vec{AE} = \frac{1}{4} \vec{AB} = \frac{\vec{b}}{4}$$

$$\vec{CE} = \vec{AE} - \vec{AC} = \frac{\vec{b}}{4} - \vec{c}$$

$$\cos \theta = \frac{\vec{CE} \cdot \vec{CA}}{|\vec{CE}| |\vec{CA}|}$$

$$\cos \theta = \frac{\left(\frac{\vec{b}}{4} - \vec{c}\right) \cdot (-\vec{c})}{\left|\frac{\vec{b}}{4} - \vec{c}\right| |\vec{c}|}$$

$$= \frac{\vec{c} \cdot \vec{c} - \frac{\vec{b} \cdot \vec{c}}{4}}{\left|\frac{\vec{b}}{4} - \vec{c}\right| |\vec{c}|} = \frac{3\sqrt{7}}{8}$$

W.E-7: If the distance of the point $B(\vec{i} + 2\vec{j} + 3\vec{k})$ from the line which is passing through

$A(4\vec{i} + 2\vec{j} + 2\vec{k})$ and Parallel to the vector

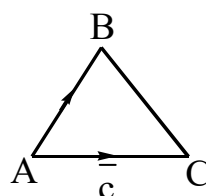
$\vec{c} = 2\vec{i} + 3\vec{j} + 6\vec{k}$ is λ then $\lambda^6 + \lambda^4 + \lambda^2 + 1$ is
A. 1001 B. 1101 C. 1111 D. 1011

Sol: (D)

Given $B = (1, 2, 3)$

$A = (4, 2, 2)$ and

$$\vec{c} = 2\vec{i} + 3\vec{j} + 6\vec{k}$$



$$\text{Then } \lambda = \frac{|\vec{AB} \times \vec{c}|}{|\vec{c}|} = \sqrt{10}$$

$$\therefore \lambda^6 + \lambda^4 + \lambda^2 + 1 = 1000 + 100 + 10 + 1 = 1111$$

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W.E-8: $\vec{a} = -\hat{i} - \hat{k}$, $\vec{b} = -\hat{i} + \hat{j}$ and $\vec{c} = \hat{i} + 2\hat{j} + 3\hat{k}$ be three given vectors. if $\vec{r}$ is a vector such

that $\vec{r} \times \vec{b} = \vec{c} \times \vec{b}$ and $\vec{r} \cdot \vec{a} = 0$ then the value of $\vec{r} \cdot \vec{b}$ is

Sol: (9)

$$\text{Given } \vec{r} \times \vec{b} = \vec{c} \times \vec{b}$$

Taking cross product on both sides with $\vec{a}$

$$\Rightarrow (\vec{r} \times \vec{b}) \times \vec{a} = (\vec{c} \times \vec{b}) \times \vec{a}$$

$$\Rightarrow (\vec{r} \cdot \vec{a})\vec{b} - (\vec{a} \cdot \vec{b})\vec{r} = (\vec{c} \cdot \vec{a})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$$

$$\Rightarrow \vec{0} - \vec{r} = -4\vec{b} - \vec{c}$$

$$\Rightarrow \vec{r} = 4\vec{b} + \vec{c}$$

$$\Rightarrow \vec{r} = -3\hat{i} + 6\hat{j} + 3\hat{k}$$

$$\therefore \vec{r} \cdot \vec{b} = 3 + 6 = 9$$

W.E-9: Let O be an interior point of ΔABC such that $\vec{OA} + 2\vec{OB} + 3\vec{OC} = \vec{0}$, Then the ratio of the area of ΔABC to the area of ΔAOC is

Sol: (3)

$$\frac{\text{Area of } \Delta ABC}{\text{Area of } \Delta AOC} = \frac{(1/2)|\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}|}{(1/2)|\vec{a} \times \vec{b}|}$$

$$\text{Now } \vec{a} + 2\vec{b} + 3\vec{c} = \vec{0}$$

$$\text{cross with } \vec{b}, \vec{a} \times \vec{b} + 3\vec{c} \times \vec{b} = \vec{0}$$

$$\Rightarrow \vec{a}, 2\vec{a} \times \vec{b} + 3\vec{a} \times \vec{c} = \vec{0}$$

$$\Rightarrow \vec{a} \times \vec{b} = 3(\vec{b} \times \vec{c})$$

$$\text{Hence, } \vec{a} \times \vec{b} = \frac{3}{2}(\vec{c} \times \vec{a}) = 3(\vec{b} \times \vec{c})$$

Let $(\vec{c} \times \vec{a}) = \vec{p}$. Then

$$\vec{a} \times \vec{b} = \frac{3\vec{p}}{2} \quad \vec{b} \times \vec{c} = \frac{\vec{p}}{2}$$

Hence, the ratio is

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$$\begin{aligned} &= \frac{|\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}|}{|\vec{p}|} \\ &= \frac{|(3\vec{p}/2) + (\vec{p}/2) + \vec{p}|}{|\vec{p}|} \\ &= \frac{3|\vec{p}|}{|\vec{p}|} = 3 \end{aligned}$$

W.E-10: If the planes $x - cy - bz = 0$, $cx - y + az = 0$ and $bx + ay - z = 0$ Pass through a line then the value of $a^2 + b^2 + c^2 + 2abc$ is

Sol: (1)

Given plane are

$$x - cy - bz = 0 \dots (1)$$

$$cx - y + az = 0 \dots (2)$$

$$bx + ay - z = 0 \dots (3)$$

Equation of plane passing through the line of intersection of planes (1) and (2) may be taken as

$$(x - cy - bz) + \lambda(cx - y + az) = 0 \dots (4)$$

Now, Planes (3) and (4) are same

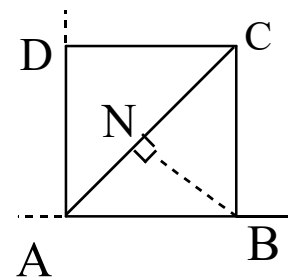
$$\therefore \frac{1 + c\lambda}{b} = \frac{-(c + \lambda)}{a} = \frac{-b + a\lambda}{-1}$$

By Eliminating λ , we get $a^2 + b^2 + c^2 + 2abc = 1$

W.E-11: ABCD is a square of unit side. It is folded along the diagonal AC, So that the plane ABC is Perpendicular to the plane ACD. The shortest distance between AB and CD is

$$\text{A) } \sqrt{3} \quad \text{B) } \frac{1}{\sqrt{3}} \quad \text{C) } \frac{\sqrt{3}}{2} \quad \text{D) } \frac{\sqrt{2}}{\sqrt{3}}$$

Sol: (D)



Let Square ABCD initially lies in xy-plane with A lying at Origin, $\vec{AB}$ along x-axis and AD along y-axis then $AB = BC = CD = AD = 1$

Let 'N' be the foot of perpendicular from 'B' to

$$AC \text{ then } BN = \frac{1}{\sqrt{2}}$$

Let B' be new position of B after folding along diagonal AC then Co-ordinates of various points in 3D are

$$A=(0,0,0), B=(1,0,0), C=(1,1,0), D=(0,1,0)$$

$$B' = \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{\sqrt{2}}\right), N = \left(\frac{1}{2}, \frac{1}{2}, 0\right)$$

Now $\overrightarrow{AB}$ line equation is

$$\vec{r} = t \left(\frac{1}{2}\vec{i} + \frac{1}{2}\vec{j} + \frac{1}{\sqrt{2}}\vec{k} \right)$$

$$\overrightarrow{CD} \text{ line equation is } \vec{r} = (i+j) + s(-i)$$

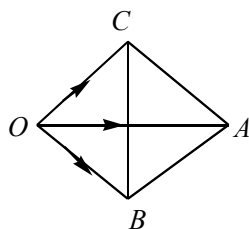
$\therefore$ shortest distance

$$= \frac{|\begin{vmatrix} \vec{a}-\vec{c} & \vec{b}-\vec{d} \\ \vec{b} \times \vec{d} \end{vmatrix}|}{|\vec{b} \times \vec{d}|} = \frac{\begin{vmatrix} 1 & 1 & 0 \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{\sqrt{2}} \\ -1 & 0 & 0 \end{vmatrix}}{\begin{vmatrix} -\frac{1}{2}\vec{k} + \frac{1}{2}\vec{j} \end{vmatrix}} = \sqrt{\frac{2}{3}}$$

W.E-12: The length of two opposite edges of a tetrahedron are a, b and their shortest distance is d and angle between them is θ then volume of tetrahedron is

- A. $\frac{1}{6}abd \sin \theta$ B. $\frac{2}{6}abd \sin \theta$
C. $d \sin \theta$ D. $\frac{1}{6}abd \cos \theta$

Sol: (A)



Let $\overrightarrow{OA} = \vec{l}, \overrightarrow{OB} = \vec{m}, \overrightarrow{OC} = \vec{n}$ then

The Equation of line $\overrightarrow{OA}$ is $\vec{r} = t\vec{l}$(1)

and Equation of line $\overrightarrow{BC}$ is

$$\vec{r} = \vec{m} + s(\vec{n} - \vec{m}) \text{.....(2)}$$

Since $\overrightarrow{OA}, \overrightarrow{BC}$ are opposite edges

Now *shortest distance* = d

$$\Rightarrow \frac{|\begin{vmatrix} \vec{m} & \vec{l} & \vec{n}-\vec{m} \end{vmatrix}|}{|\vec{l} \times (\vec{n}-\vec{m})|} = d \Rightarrow \frac{-[\vec{l} \vec{m} \vec{n}]}{ab \sin \theta} = d$$

$$\Rightarrow [\vec{l} \vec{m} \vec{n}] = -abd \sin \theta$$

$$\therefore \text{volume} = \frac{1}{6}[\vec{l} \vec{m} \vec{n}] = \frac{1}{6}abd \sin \theta$$

W.E-13: Let $\vec{a} = \vec{i} + 2\vec{j} + 2\vec{k}, \vec{b} = 2\vec{i} + 3\vec{j} + 6\vec{k}$ and

$\vec{c} = 4\vec{i} + 4\vec{j} + 7\vec{k}$. A vector which is equally inclined to these three vectors is perpendicular to the plane passing through the points $p\vec{a}, q\vec{b}$ and $r\vec{c}$. If p, q, r are the least possible positive integers, then the value of $p+q+r$ is

- A)19 B)37 C)53 D)111

Sol: (B)

Let $\vec{r}$ be required vector then

$$(\vec{r}, \vec{a}) = (\vec{r}, \vec{b}) = (\vec{r}, \vec{c})$$

$$\cos(\vec{r}, \vec{a}) = \cos(\vec{r}, \vec{b}) = \cos(\vec{r}, \vec{c})$$

$$\Rightarrow \frac{x+2y+2z}{3} = \frac{2x+3y+6z}{7} = \frac{4x+4y+7z}{9} = k(\text{say})$$

$$\therefore x+2y+2z = 3k \text{.....(1)}$$

$$2x+3y+6z = 7k \text{.....(2)}$$

$$4x+4y+7z = 9k \text{.....(3)}$$

by solving above equations we get

$$x:y:z = 3:5:7$$

$$\therefore \vec{r} = 3\vec{i} + 5\vec{j} + 7\vec{k}$$

Since $\vec{r}$ is perpendicular to the plane passing through the points $p\vec{a}, p\vec{b}, r\vec{c}$

$$\therefore \vec{r} \cdot (p\vec{a} - p\vec{b}) = 0, \vec{r} \cdot (q\vec{b} - r\vec{c}) = 0$$

$$\Rightarrow p(\vec{r} \cdot \vec{a}) = q(\vec{r} \cdot \vec{b}) \text{ and } q(\vec{r} \cdot \vec{b}) = r(\vec{r} \cdot \vec{c})$$

$$\Rightarrow p(27) = q(63) = r(81)$$

$$\Rightarrow \frac{p}{21} = \frac{q}{9} = \frac{r}{7}$$

$$\therefore \text{least possible value of } p+q+r = 37$$

W.E-14: Let the position vector of the orthocentre of $\triangle ABC$ be $\vec{r}$, then which of the following statement(s) is/are correct (Given position

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vectors of points A,B,C are $\hat{a}, \hat{b}, \hat{c}$ and $abc \neq 0$)

$$A. \hat{r} \cdot \hat{i} = \frac{a}{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}}$$

$$B. \hat{r} \cdot \hat{i} = \frac{1}{a \left(\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \right)}$$

$$C. \frac{\hat{r} \cdot \hat{i}}{\hat{r} \cdot \hat{j}} + \frac{\hat{r} \cdot \hat{j}}{\hat{r} \cdot \hat{k}} + \frac{\hat{r} \cdot \hat{k}}{\hat{r} \cdot \hat{i}} = \frac{a}{b} + \frac{b}{c} + \frac{c}{a}$$

$$D. \frac{\hat{r} \cdot \hat{i}}{\hat{r} \cdot \hat{j}} + \frac{\hat{r} \cdot \hat{j}}{\hat{r} \cdot \hat{k}} + \frac{\hat{r} \cdot \hat{k}}{\hat{r} \cdot \hat{i}} = \frac{b}{a} + \frac{c}{b} + \frac{a}{c}$$

Sol: (B,D)

Let orthocentre $\vec{r} = \alpha \hat{i} + \beta \hat{j} + \gamma \hat{k}$

So $a\alpha = b\beta = c\gamma$

$$\Rightarrow \alpha = \frac{\lambda}{a}, \beta = \frac{\lambda}{b}, \gamma = \frac{\lambda}{c}$$

And point (α, β, γ) is on plane $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

$$\text{So } \frac{\lambda}{a^2} + \frac{\lambda}{b^2} + \frac{\lambda}{c^2} = 1 \Rightarrow \lambda = \frac{1}{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}}$$

$$\text{So, } \alpha = \frac{1}{a \left(\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \right)}, \beta = \frac{1}{b \left(\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \right)}, \gamma = \frac{1}{c \left(\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \right)}$$

W.E-15: Let $\vec{a}, \vec{b}, \vec{c}$ are three vectors along the adjacent edges of a tetrahedron, if $|\vec{a}| = |\vec{b}| = |\vec{c}| = 2$ and $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = 1$, then volume of tetrahedron is

$$A. \frac{1}{\sqrt{2}} \quad B. \frac{2}{\sqrt{3}} \quad C. \sqrt{\frac{3}{2}} \quad D. \frac{2\sqrt{2}}{3}$$

Sol: (C)
volume =

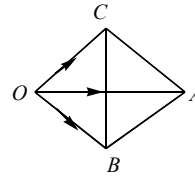
$$\frac{1}{6} \sqrt{\begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix}} = \frac{1}{6} \sqrt{\begin{vmatrix} 4 & 1 & 1 \\ 1 & 4 & 1 \\ 1 & 1 & 4 \end{vmatrix}} = \sqrt{\frac{3}{2}}$$

W.E-16. The shortest distance between the two opposite edges of a regular tetrahedron of

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edge $\sqrt{8}$ units is

sol: (2)



Let $\vec{OA} = \vec{l}, \vec{OB} = \vec{m}, \vec{OC} = \vec{n}$

The Equation of line $\vec{OA}$ is $\vec{r} = t\vec{l}$(1)

and Equation of line $\vec{BC}$ is

$$\vec{r} = \vec{m} + s(\vec{n} - \vec{m}) \dots \dots \dots (2)$$

Since $\vec{OA}, \vec{BC}$ are opposite edges

$$\text{Now, S.D} = \frac{|\vec{m} \times \vec{l} \cdot \vec{n} - \vec{m}|}{|\vec{l} \times (\vec{n} - \vec{m})|}$$

$$= \frac{|\vec{l} \cdot \vec{m} \times \vec{n}|}{|\vec{l} \times (\vec{n} - \vec{m})|} = \frac{16}{8} = 2$$

W.E-17: $\vec{a}, \vec{b}, \vec{c}$ are three unit vectors equally inclined

to each other at an angle $\frac{\pi}{3}$ then the value

$$\text{of } [\vec{a} + \vec{b} \quad \vec{b} + \vec{c} \quad \vec{c} + \vec{a}]^2 + [\vec{a} \times \vec{b} \quad \vec{b} \times \vec{c} \quad \vec{c} \times \vec{a}] \text{ is}$$

$$A. \frac{3}{4} \quad B. \frac{5}{2} \quad C. \frac{7}{2} \quad D. 4$$

Sol: (B)

$$[\vec{a} \quad \vec{b} \quad \vec{c}]^2 = \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix} = \begin{vmatrix} 1 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 1 \end{vmatrix} = \frac{1}{2}$$

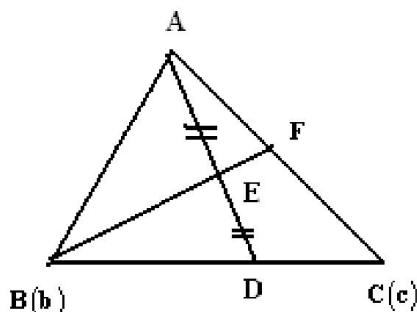
$$\therefore [\vec{a} + \vec{b} \quad \vec{b} + \vec{c} \quad \vec{c} + \vec{a}]^2 + [\vec{a} \times \vec{b} \quad \vec{b} \times \vec{c} \quad \vec{c} \times \vec{a}] =$$

$$\left(2[\vec{a} \quad \vec{b} \quad \vec{c}] \right)^2 + [\vec{a} \quad \vec{b} \quad \vec{c}]^2 = \frac{5}{2}$$

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W.E-18: The median AD of the triangle ABC is bisected at E and BE meets AC at F. If $AF:AC = 1/k$ then the value k



- (A) 1 (B) 2 (C) 3 (D) 4
sol: (C)

By using menelaw's theorem $\frac{AE}{ED} \cdot \frac{DB}{BC} \cdot \frac{CF}{FA} = 1$

$$\Rightarrow \frac{1}{1} \cdot \frac{1}{2} \cdot \frac{CF}{FA} = 1 \Rightarrow \frac{AF}{FC} = \frac{1}{2}$$

$$\therefore \frac{AF}{AC} = \frac{1}{3} \Rightarrow K = 3$$

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SINGLE ANSWER QUESTIONS

- ABCD is a quadrilateral. E is the point of intersection of the line joining the midpoints of the opposite sides. If O is any point and $\overrightarrow{OA} + \overrightarrow{OB} + \overrightarrow{OC} + \overrightarrow{OD} = x\overrightarrow{OE}$ then x is equal to
(A) 3 (B) 9 (C) 7 (D) 4
- 'I' is the incentre of triangle ABC, whose corresponding sides are a, b, c respectively. $a\overrightarrow{IA} + b\overrightarrow{IB} + c\overrightarrow{IC}$ is always equal to
(A) $\vec{0}$ (B) $(a+b+c)\overrightarrow{BC}$
(C) $(\vec{a} + \vec{b} + \vec{c})\overrightarrow{AC}$ (D) $(a+b+c)\overrightarrow{AB}$
- If D, E, F are midpoints of the sides BC, CA, AB respectively of a triangle ABC and O is any point then $\overrightarrow{AD} + \frac{2}{3}\overrightarrow{BE} + \frac{1}{3}\overrightarrow{CF} =$
(A) $\overrightarrow{AC}$ (B) $2\overrightarrow{AC}$ (C) $\frac{1}{2}\overrightarrow{AC}$ (D) $\vec{0}$

- If $\vec{a}$ and $\vec{b}$ are non-collinear vectors such that $\vec{p} = (x+4)\vec{a} + \vec{b}$ and $\vec{q} = (2-2x)\vec{a} + 2\vec{b}$ are collinear, then the value of x is
(A) $3/2$ (B) $-3/2$ (C) $2/3$ (D) $-2/3$

- Let 'O' be the centre of regular pentagon and $\overrightarrow{OA} = \vec{a}$ then

$$\overrightarrow{AB} + 2\overrightarrow{BC} + 3\overrightarrow{CD} + 4\overrightarrow{DE} + 5\overrightarrow{EA} =$$

- (A) $\vec{0}$ (B) $4\vec{a}$ (C) $5\vec{a}$ (D) $15\vec{a}$

- If in ΔABC , $\overrightarrow{AB} = \frac{\vec{u}}{|\vec{u}|} - \frac{\vec{v}}{|\vec{v}|}$ and $\overrightarrow{AC} = \frac{2\vec{u}}{|\vec{u}|}$

where $|\vec{u}| \neq |\vec{v}|$ then

- (A) $1 + \cos 2A + \cos 2B + \cos 2C = 0$
(B) $1 + \cos 2A + \cos 2B + \cos 2C = 2$
(C) Both A and B
(D) None of these

- If $\vec{a}$ and $\vec{b}$ are two unit vectors and θ is the angle between them, then the unit vector along the angular bisector of $\vec{a}$ and $\vec{b}$ will given by

- (A) $\frac{\vec{a} - \vec{b}}{2 \cos \theta / 2}$ (B) $\frac{\vec{a} + \vec{b}}{2 \cos \theta / 2}$
(C) $\frac{\vec{a} - \vec{b}}{\cos \theta / 2}$ (D) $\frac{\vec{a} + \vec{b}}{\cos \theta / 2}$

- $\vec{a}$ and $\vec{c}$ are unit vectors and $|\vec{b}| = 4$. The

angle between $\vec{a}$ and $\vec{c}$ is $\cos^{-1}\left(\frac{1}{4}\right)$. If

$\vec{b} - 2\vec{c} = \lambda\vec{a}$ then value of λ is

- A) 3, -4 B) $\frac{1}{4}, \frac{3}{4}$ C) -3, 4 D) $-\frac{1}{4}, \frac{3}{4}$

- Let $\vec{u}, \vec{v}$ and $\vec{w}$ be such that $|\vec{u}| = 1, |\vec{v}| = 2$ and $|\vec{w}| = 3$. If the projection of $\vec{v}$ along $\vec{u}$ is equal to that of $\vec{w}$ along $\vec{u}$ and vectors $\vec{v}$

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and $\vec{w}$ are perpendicular to each other, then $|\vec{u} - \vec{v} + \vec{w}|$ equals to

- A) 2 B) $\sqrt{7}$ C) $\sqrt{14}$ D) 14

10. Let $\vec{u}, \vec{v}$ and $\vec{w}$ vectors such that $\vec{u} + \vec{v} + \vec{w} = \vec{0}$. If $|\vec{u}| = 3, |\vec{v}| = 4$ and $|\vec{w}| = 5$, then $\vec{u} \cdot \vec{v} + \vec{v} \cdot \vec{w} + \vec{w} \cdot \vec{u}$ is [IIT JEE 1995]
(A) 47 (B) -25 (C) 0 (D) 25

11. If $\vec{a}, \vec{b}$ and $\vec{c}$ are unit vectors, then $|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2$ does not exceed [IIT JEE 2001]
(A) 4 (B) 9 (C) 8 (D) 6

12. If $\vec{a}$ and $\vec{b}$ are two unit vectors such that $\vec{a} + 2\vec{b}$ and $5\vec{a} - \vec{b}$ are perpendicular to each other, then the angle between $\vec{a}$ and $\vec{b}$ is [IIT JEE 2002]
(A) 45° (B) 60°
(C) $\cos^{-1}\left(\frac{1}{3}\right)$ (D) $\cos^{-1}\left(\frac{2}{7}\right)$

13. 4-points whose position Vectors $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ are coplanar and $(\sin \alpha)\vec{a} + (2 \sin 2\beta)\vec{b} + (3 \sin 3\gamma)\vec{c} - \vec{d} = \vec{0}$. then the least value of $\sin^2 \alpha + \sin^2 2\beta + \sin^2 3\gamma$ is

- A) $\frac{1}{14}$ B) 14 C) 6 D) $\frac{1}{\sqrt{6}}$

14. If a is a real constant and A, B and C are variable angles and $\sqrt{a^2 - 4} \tan A + a \tan B + \sqrt{a^2 + 4} \tan C = 6a$, then the least value of $\tan^2 A + \tan^2 B + \tan^2 C$ is
(A) 6 (B) 10 (C) 12 (D) 3

15. Let $\hat{a}$ and $\hat{b}$ be mutually perpendicular unit vectors. If $\vec{r}$ is any arbitrary vector then
A) $\vec{r} = (\vec{r} \cdot \hat{a})\hat{a} + (\vec{r} \cdot \hat{b})\hat{b} + (\vec{r} \cdot (\hat{a} \times \hat{b}))(\hat{a} \times \hat{b})$
B) $\vec{r} = (\vec{r} \cdot \hat{a})\hat{a} - (\vec{r} \cdot \hat{b})\hat{b} - (\vec{r} \cdot (\hat{a} \times \hat{b}))(\hat{a} \times \hat{b})$

C) $\vec{r} = (\vec{r} \cdot \hat{a})\hat{a} - (\vec{r} \cdot \hat{b})\hat{b} + (\vec{r} \cdot (\hat{a} \times \hat{b}))(\hat{a} \times \hat{b})$

D) $\vec{r} = (\vec{r} \cdot \hat{a})\hat{a} + (\vec{r} \cdot \hat{b})\hat{b}$

16. Let $\vec{OA} = \vec{i} + 2\vec{j} + 2\vec{k}$. In the plane $\vec{OA}$ and $\vec{i}$ rotate $\vec{OA}$ through 90° about the origin O such that the new position of $\vec{OA}$ is

A) $\frac{1}{\sqrt{2}}(4\vec{i} - \vec{j} - \vec{k})$ B) $\frac{1}{\sqrt{2}}(-4\vec{i} + 2\vec{j} + 2\vec{k})$

C) $2\vec{j} - 2\vec{k}$ D) $6\vec{i} - 3\vec{k}$

17. Let two non-collinear unit vectors $\hat{a}$ and $\hat{b}$ form an acute angle. A point P moves so that at any time t the position vector $\vec{OP}$ (where O is the origin) is given by $\hat{a} \cos t + \hat{b} \sin t$. When P is farthest from origin O, let M be the length of $\vec{OP}$ and $\hat{u}$ be the unit vector along $\vec{OP}$ then [IIT JEE 2008]

(A) $\hat{u} = \frac{\hat{a} + \hat{b}}{|\hat{a} + \hat{b}|}$ and $M = (1 + \hat{a} \cdot \hat{b})^{1/2}$

(B) $\hat{u} = \frac{\hat{a} - \hat{b}}{|\hat{a} - \hat{b}|}$ and $M = (1 + \hat{a} \cdot \hat{b})^{1/2}$

(C) $\hat{u} = \frac{\hat{a} + \hat{b}}{|\hat{a} + \hat{b}|}$ and $M = (1 + 2\hat{a} \cdot \hat{b})^{1/2}$

(D) $\hat{u} = \frac{\hat{a} - \hat{b}}{|\hat{a} - \hat{b}|}$ and $M = (1 + 2\hat{a} \cdot \hat{b})^{1/2}$

18. If $\vec{a}$ satisfies $\vec{a} \times (\hat{i} + 2\hat{j} + \hat{k}) = \hat{i} - \hat{k}$ then $\vec{a}$ is equal to

A) $\lambda \hat{i} + (2\lambda - 1)\hat{j} + \lambda \hat{k}, \lambda \in R$

B) $\lambda \hat{i} + (1 - 2\lambda)\hat{j} + \lambda \hat{k}, \lambda \in R$

C) $\lambda \hat{i} + (2\lambda + 1)\hat{j} + \lambda \hat{k}, \lambda \in R$

D) $\lambda \hat{i} - (1 + 2\lambda)\hat{j} + \lambda \hat{k}, \lambda \in R$

19. Let the position vectors of the points P and Q be $4\hat{i} + \hat{j} + \lambda \hat{k}$ and $2\hat{i} - \hat{j} + \lambda \hat{k}$ respectively. Vector $\hat{i} - \hat{j} + 6\hat{k}$ is

perpendicular to the plane containing the origin and the points P and Q then λ is

- A) $-\frac{1}{2}$ B) $\frac{1}{2}$ C) 1 D) -1

20. If $\vec{a}, \vec{b}, \vec{c}$ are unit vectors such that $\vec{a} \cdot \vec{b} = 0 = \vec{a} \cdot \vec{c}$ and the angle between $\vec{b}$ and $\vec{c}$ is $\frac{\pi}{3}$ then the value of $|\vec{a} \times \vec{b} - \vec{a} \times \vec{c}|$ is

- A) $\frac{1}{2}$ B) 1 C) 2 D) $\frac{1}{3}$

21. If $\vec{a}$ and $\vec{b}$ are any two vectors of magnitudes 1 and 2 respectively, and $(1 - 3\vec{a} \cdot \vec{b})^2 + |2\vec{a} + \vec{b} + 3(\vec{a} \times \vec{b})|^2 = 47$ then the angle between $\vec{a}$ and $\vec{b}$ is

- A) $\frac{\pi}{3}$ B) $\pi - \cos^{-1}\left(\frac{1}{4}\right)$
C) $\frac{2\pi}{3}$ D) $\cos^{-1}\left(\frac{1}{4}\right)$

22. If $\vec{a}$ and $\vec{b}$ are any two vectors of magnitudes 2 and 3 respectively such that $|2(\vec{a} \times \vec{b})| + |3(\vec{a} \cdot \vec{b})| = k$ then the maximum value of k is

- A) $\sqrt{13}$ B) $2\sqrt{13}$ C) $6\sqrt{13}$ D) $10\sqrt{13}$

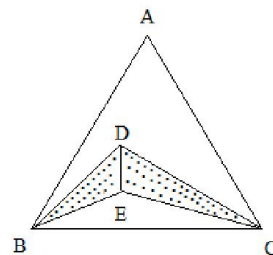
23. The vertex A of triangle ABC is on the line $\vec{r} = \hat{i} + \hat{j} + \lambda \hat{k}$ and the vertices B and C have respective position vectors $\hat{i}$ and $\hat{j}$. Let Δ be the area of the triangle and

$\Delta \in \left[\frac{3}{2}, \frac{\sqrt{33}}{2}\right]$ then the range of values of

λ corresponding to Δ is

- A) $[-8, -4] \cup [4, 8]$ B) $[-4, 4]$
C) $[-2, 2]$ D) $[-4, -2] \cup [2, 4]$

24. In figure, $\vec{AB} = 3\vec{i} - \vec{j}$, $\vec{AC} = 2\vec{i} + 3\vec{j}$ and $\vec{DE} = 4\vec{i} - 2\vec{j}$ then the area of the shaded region in square units is



- A) 5 B) 6 C) 7 D) 8

25. If $\vec{a} = (\hat{i} + \hat{j} + \hat{k})$, $\vec{a} \cdot \vec{b} = 1$ and $\vec{a} \times \vec{b} = \hat{j} - \hat{k}$ then $\vec{b}$ is [IIT JEE 2003]

- (A) $\hat{i} - \hat{j} + \hat{k}$ (B) $2\hat{j} - \hat{k}$
(C) $\hat{i}$ (D) $2\hat{i}$

26. Let $\vec{a}, \vec{b}, \vec{c}$ be unit vectors such that $\vec{a} + \vec{b} + \vec{c} = \vec{0}$. Which one of the following is correct [IIT JEE 2007]

(A) $\vec{a} \times \vec{b} \neq \vec{b} \times \vec{c} = \vec{c} \times \vec{a} = \vec{0}$

(B) $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a} \neq \vec{0}$

(C) $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{a} \times \vec{c} = \vec{0}$

(D) $\vec{a} \times \vec{b}, \vec{b} \times \vec{c}, \vec{c} \times \vec{a}$ are mutually perpendicular

27. If $\vec{a}$ and $\vec{b}$ are vectors such that $|\vec{a} + \vec{b}| = \sqrt{29}$ and $\vec{a} \times (2\hat{i} + 3\hat{j} + 4\hat{k}) = (2\hat{i} + 3\hat{j} + 4\hat{k}) \times \vec{b}$ then a possible value of $(\vec{a} + \vec{b}) \cdot (-7\hat{i} + 2\hat{j} + 3\hat{k})$ is [IITJEE 2012]

- A) 0 B) 3 C) 4 D) 8

28. If the vectors $\vec{a}, \vec{b}$ and $\vec{c}$ form the sides BC, CA and AB respectively of a triangle ABC then [IIT JEE 2000]

(A) $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = 0$ (B) $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a}$

(C) $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a}$ (D) $\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a} = \vec{0}$

29. Let ΔABC be a given triangle. If

$|\vec{BA} - t\vec{BC}| \geq |\vec{AC}|$ for any $t \in R$ then

ΔABC is

(A) Equilateral

(B) Right angled

(C) Isosceles

(D) None of these

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30. $\vec{a}$ and $\vec{b}$ are two mutually perpendicular unit vectors and $\vec{c}$ is a unit vector inclined at an angle θ to both $\vec{a}$ and $\vec{b}$ if $\vec{c} = x\vec{a} + y\vec{b} + z(\vec{a} \times \vec{b})$, where $x, y \in R$, exhaustive range of θ is

- A) $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$ B) $\left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$
C) $\left[\pi, \frac{\pi}{2}\right]$ D) $\left[0, \frac{\pi}{2}\right]$

31. $\vec{b}$ and $\vec{c}$ are unit vectors. Then for any arbitrary vector $\vec{a}$ the value of

$$\left(\left((\vec{a} \times \vec{b}) + (\vec{a} \times \vec{c})\right) \times (\vec{b} \times \vec{c})\right) \cdot (\vec{b} - \vec{c})$$

is always equal to

- A) $|\vec{a}|$ B) $\frac{1}{2}|\vec{a}|$ C) $\frac{1}{3}|\vec{a}|$ D) 0

32. If $\vec{a}$ and $\vec{b}$ are non-zero non-collinear vectors then

$$[\vec{a} \vec{b} \hat{i}] \hat{i} + [\vec{a} \vec{b} \hat{j}] \hat{j} + [\vec{a} \vec{b} \hat{k}] \hat{k} \text{ is equal to}$$

- A) $\vec{a} + \vec{b}$ B) $\vec{a} \times \vec{b}$ C) $\vec{a} - \vec{b}$ D) $\vec{b} \times \vec{a}$

33. If $\vec{a} = 2\hat{i} + \hat{j} + \hat{k}$, $\vec{b} = \hat{i} + 2\hat{j} + 2\hat{k}$, $\vec{c} = \hat{i} + \hat{j} + 2\hat{k}$ and $(1+\alpha)\hat{i} + \beta(1+\alpha)\hat{j} + \gamma(1+\alpha)(1+\beta)\hat{k} = \vec{a} \times (\vec{b} \times \vec{c})$ then α, β and γ are

- A) $-2, -4, -\frac{2}{3}$ B) $2, -4, \frac{2}{3}$
C) $-2, 4, \frac{2}{3}$ D) $2, 4, -\frac{2}{3}$

34. Let a, b, c be distinct non-negative numbers. If the vectors $a\hat{i} + a\hat{j} + c\hat{k}$, $\hat{i} + \hat{k}$ and $c\hat{i} + c\hat{j} + b\hat{k}$ lie in a plane then c is

[IIT JEE 1993]

- (A) the Arithmetic Mean of a and b
(B) the Geometric Mean of a and b
(C) the Harmonic Mean of a and b
(D) equal to zero

35. Let $\vec{a} = \hat{i} - \hat{j}$, $\vec{b} = \hat{j} - \hat{k}$, $\vec{c} = \hat{k} - \hat{i}$. If $\vec{d}$ is a unit vector such that $\vec{a} \cdot \vec{d} = 0 = [\vec{b} \vec{c} \vec{d}]$ then $\vec{d}$ equals to [IIT JEE 1995]

- (A) $\pm \frac{\hat{i} + \hat{j} - 2\hat{k}}{\sqrt{6}}$ (B) $\pm \frac{\hat{i} + \hat{j} - \hat{k}}{\sqrt{3}}$
(C) $\pm \frac{\hat{i} + \hat{j} + \hat{k}}{\sqrt{3}}$ (D) $\pm \hat{k}$

36. If $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = 4\hat{i} + 3\hat{j} + 4\hat{k}$ and $\vec{c} = \hat{i} + \alpha\hat{j} + \beta\hat{k}$ are linearly dependent vectors and $|\vec{c}| = \sqrt{3}$ then [IIT JEE 1998]

- (A) $\alpha = 1, \beta = -1$ (B) $\alpha = 1, \beta = \pm 1$
(C) $\alpha = -1, \beta = \pm 1$ (D) $\alpha = \pm 1, \beta = 1$

37. For three vectors $\vec{u}, \vec{v}, \vec{w}$ which of the following expressions is not equal to any of the remaining three? [IIT JEE 1998]

- (A) $\vec{u} \cdot (\vec{v} \times \vec{w})$ (B) $(\vec{v} \times \vec{w}) \cdot \vec{u}$
(C) $\vec{v} \cdot (\vec{u} \times \vec{w})$ (D) $(\vec{u} \times \vec{v}) \cdot \vec{w}$

38. Let $\vec{a} = 2\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{b} = \hat{i} + \hat{j}$. If $\vec{c}$ is a vector such that $\vec{a} \cdot \vec{c} = |\vec{c}|$, $|\vec{c} - \vec{a}| = 2\sqrt{2}$ and the angle between $(\vec{a} \times \vec{b})$ and $\vec{c}$ is 30° then $|(\vec{a} \times \vec{b}) \times \vec{c}|$ is equal to [IIT JEE 1999]

- (A) $\frac{2}{3}$ (B) $\frac{3}{2}$ (C) 2 (D) 3

39. Let $\vec{a} = 2\hat{i} + \hat{j} + \hat{k}$, $\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$ and a unit vector $\vec{c}$ be coplanar. If $\vec{c}$ is perpendicular to $\vec{a}$ then $\vec{c}$ is equal to [IIT JEE 1999]

- (A) $\frac{1}{\sqrt{2}}(-\hat{j} + \hat{k})$ (B) $\frac{1}{\sqrt{3}}(-\hat{i} - \hat{j} - \hat{k})$
(C) $\frac{1}{\sqrt{5}}(\hat{i} - 2\hat{j})$ (D) $\frac{1}{\sqrt{5}}(\hat{i} - \hat{j} - \hat{k})$

40. If $\vec{a}, \vec{b}$ and $\vec{c}$ are unit coplanar vectors, then the scalar triple product $[2\vec{a} - \vec{b}, 2\vec{b} - \vec{c}, 2\vec{c} - \vec{a}]$ is equal to [IIT JEE 2000]

- (A) 0 (B) 1 (C) $-\sqrt{3}$ (D) $\sqrt{3}$

41. Let $\vec{a} = \hat{i} - \hat{k}$, $\vec{b} = x\hat{i} + \hat{j} + (1-x)\hat{k}$ and

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$\vec{c} = y\hat{i} + x\hat{j} + (1+x-y)\hat{k}$ then $[\vec{a} \ \vec{b} \ \vec{c}]$ depends on [IIT JEE 2001]

- (A) only x (B) only y
(C) neither x nor y (D) both x and y

42. Let $\vec{V} = 2\hat{i} + \hat{j} - \hat{k}$ and $\vec{W} = \hat{i} + 3\hat{k}$. If $\vec{U}$ is a unit vector then the maximum value of the scalar triple product $[\vec{U} \ \vec{V} \ \vec{W}]$ is [IIT JEE 2002]

- (A) -1 (B) $\sqrt{10} + \sqrt{6}$
(C) $\sqrt{59}$ (D) $\sqrt{60}$

43. The value of 'a' so that the volume of parallelopiped formed by $\hat{i} + a\hat{j} + \hat{k}$, $\hat{j} + a\hat{k}$ and $a\hat{i} + \hat{k}$ become minimum, is [IIT JEE 2003]

- (A) -3 (B) 3 (C) $\frac{1}{\sqrt{3}}$ (D) $\sqrt{3}$

44. The unit vector which is orthogonal to the vector $3\hat{i} + 2\hat{j} + 6\hat{k}$ and is coplanar with the vectors $2\hat{i} + \hat{j} + \hat{k}$ and $\hat{i} - \hat{j} + \hat{k}$ is [IIT JEE 2004]

- (A) $\frac{2\hat{i} - 6\hat{j} + \hat{k}}{\sqrt{41}}$ (B) $\frac{2\hat{i} - 3\hat{j}}{\sqrt{13}}$
(C) $\frac{3\hat{j} - \hat{k}}{\sqrt{10}}$ (D) $\frac{4\hat{i} + 3\hat{j} - 3\hat{k}}{\sqrt{34}}$

45. Let $\vec{a} = \hat{i} + 2\hat{j} + \hat{k}$, $\vec{b} = \hat{i} - \hat{j} + \hat{k}$, $\vec{c} = \hat{i} + \hat{j} - \hat{k}$.

A vector coplanar to $\vec{a}$ and $\vec{b}$ has a projection along $\vec{c}$ of magnitude $\frac{1}{\sqrt{3}}$ then the vector is [IIT JEE 2006]

- (A) $4\hat{i} - \hat{j} + 4\hat{k}$ (B) $4\hat{i} + \hat{j} - 4\hat{k}$
(C) $2\hat{i} + \hat{j} + \hat{k}$ (D) none of these

46. The number of distinct real values of λ , for which the vectors $-\lambda^2\hat{i} + \hat{j} + \hat{k}$, $\hat{i} - \lambda^2\hat{j} + \hat{k}$ and $\hat{i} + \hat{j} - \lambda^2\hat{k}$ are coplanar is [IIT JEE 2006]

- (A) 0 (B) 1 (C) 2 (D) 3

47. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be the three vectors having magnitudes 1, 5 and 3 respectively, such that the angle between $\vec{a}$ and $\vec{b}$ is θ and $\vec{a} \times (\vec{a} \times \vec{b}) = \vec{c}$ then $\tan \theta$ is equal to

- A) 0 B) $\frac{2}{3}$ C) $\frac{3}{5}$ D) $\frac{3}{4}$

48. If $\vec{a} \cdot \vec{b} = \beta$ and $\vec{a} \times \vec{b} = \vec{c}$ then $\vec{b}$ is

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A) $\frac{(\beta\vec{a} - \vec{a} \times \vec{c})}{|\vec{a}|^2}$ B) $\frac{(\beta\vec{a} + \vec{a} \times \vec{c})}{|\vec{a}|^2}$

C) $\frac{(\beta\vec{c} - \vec{a} \times \vec{c})}{|\vec{a}|^2}$ D) $\frac{(\beta\vec{a} + \vec{a} \times \vec{c})}{|\vec{a}|^2}$

49. If $\vec{a} = \vec{b} + \vec{c}$, $\vec{b} \times \vec{d} = \vec{0}$ and $\vec{c} \cdot \vec{d} = 0$ then

$\frac{\vec{d} \times (\vec{a} \times \vec{d})}{|\vec{d}|^2}$ is always equal to

- A) $\vec{a}$ B) $\vec{d}$ C) $\vec{b}$ D) $\vec{c}$

50. If $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar unit vectors such

that $\vec{a} \times (\vec{b} \times \vec{c}) = \frac{(\vec{b} + \vec{c})}{\sqrt{2}}$ then the angle

between $\vec{a}$ and $\vec{b}$ is [IIT JEE 1995]

- (A) $\frac{3\pi}{4}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{2}$ (D) π

51. If $\vec{a}, \vec{b}$ and $\vec{c}$ are three non-coplanar vectors

then $(\vec{a} + \vec{b} + \vec{c}) \cdot [(\vec{a} + \vec{b}) \times (\vec{a} + \vec{c})]$ equals

[IIT JEE 1995]

- (A) 0 (B) $[\vec{a} \ \vec{b} \ \vec{c}]$
(C) $2[\vec{a} \ \vec{b} \ \vec{c}]$ (D) $-[\vec{a} \ \vec{b} \ \vec{c}]$

52. Let $\vec{p}, \vec{q}, \vec{r}$ be three mutually perpendicular vectors of the same magnitude. If a vector

$\vec{x}$ satisfies the equation $\vec{p} \times ((\vec{x} - \vec{q}) \times \vec{p}) + \vec{q} \times ((\vec{x} - \vec{r}) \times \vec{q}) + \vec{r} \times ((\vec{x} - \vec{p}) \times \vec{r}) = \vec{0}$ then $\vec{x}$ is given by [IIT JEE 1997]

- (A) $\frac{1}{2}(\vec{p} + \vec{q} - 2\vec{r})$ (B) $\frac{1}{2}(\vec{p} + \vec{q} + \vec{r})$
(C) $\frac{1}{3}(\vec{p} + \vec{q} + \vec{r})$ (D) $\frac{1}{3}(2\vec{p} + \vec{q} - \vec{r})$

53. Let the vectors $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ be such that

$(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = \vec{0}$. Let P_1 and P_2 be planes

determined by the pairs of vectors $\vec{a}, \vec{b}$ and

$\vec{c}, \vec{d}$ respectively then the angle between P_1 and P_2 is [IIT JEE 2000]

- (A) 0 (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$

54. 'OABC' is a tetrahedron whose faces are equilateral triangle of unit side such that $\vec{OA} = \vec{a}, \vec{OB} = \vec{b}$ then $3\vec{OC} =$

- (A) $\vec{a} + \vec{b} - \sqrt{2}(\vec{a} \times \vec{b})$ (B) $\vec{a} + \vec{b} - \sqrt{3}(\vec{a} \times \vec{b})$
(C) $\vec{a} + \vec{b} + \sqrt{2}(\vec{a} \times \vec{b})$ (D) $\vec{a} + \vec{b} + 2\sqrt{2}(\vec{a} \times \vec{b})$

MULTIANSWER QUESTIONS

55. The sides of a parallelogram are $2\hat{i} + 4\hat{j} - 5\hat{k}$ and $\hat{i} + 2\hat{j} + 3\hat{k}$. The unit vector parallel to one of the diagonals is

- (A) $\frac{1}{7}(3\hat{i} + 6\hat{j} - 2\hat{k})$ (B) $\frac{1}{7}(3\hat{i} - 6\hat{j} - 2\hat{k})$
(C) $\frac{1}{\sqrt{69}}(\hat{i} + 2\hat{j} + 8\hat{k})$
(D) $\frac{1}{\sqrt{69}}(-\hat{i} - 2\hat{j} + 8\hat{k})$

56. Let ABC be a triangle, the position vectors of whose vertices are $7\hat{j} + 10\hat{k}$, $-\hat{i} + 6\hat{j} + 6\hat{k}$ and $-4\hat{i} + 9\hat{j} + 6\hat{k}$. Then ΔABC is

- (A) isosceles (B) equilateral
(C) right angled (D) scalene

57. All values of λ such that $x, y, z \neq (0, 0, 0)$

$$\text{and } \left(\hat{i} + \hat{j} + 3\hat{k} \right)x + \left(3\hat{i} - 3\hat{j} + \hat{k} \right)y + \left(-4\hat{i} + 5\hat{j} \right)z = \lambda \left(x\hat{i} + y\hat{j} + z\hat{k} \right) \text{ where}$$

$\hat{i}, \hat{j}, \hat{k}$ are unit vectors along the co-ordinate axes.

- (A) 0 (B) 1 (C) -1 (D) 2

58. The vectors $\vec{a} = x\hat{i} - 2\hat{j} + 5\hat{k}$ and

$\vec{b} = \hat{i} + y\hat{j} - z\hat{k}$ are collinear if

- (A) $x=1, y=-2, z=-5$
(B) $x=1/2, y=-4, z=-10$
(C) $x=-1/2, y=4, z=10$
(D) $x=-1, y=2, z=5$

59. A vector of magnitude 2 along a bisector of the angle between the two vectors

$2\hat{i} - 2\hat{j} + \hat{k}$ and $\hat{i} + 2\hat{j} - 2\hat{k}$ is

- (A) $\frac{2}{\sqrt{10}}(3\hat{i} - \hat{k})$ (B) $\frac{1}{\sqrt{26}}(\hat{i} - 4\hat{j} + 3\hat{k})$
(C) $\frac{2}{\sqrt{26}}(\hat{i} - 4\hat{j} + 3\hat{k})$ (D) $\frac{1}{\sqrt{10}}(\hat{i} + 3\hat{k})$

60. If A(-4,0,3), B(14,2,-5) then which one of the following points lie on the bisector of the angle between $\vec{OA}$ and $\vec{OB}$ (O is the origin of reference)

- (A) (2,2,4) (B) (2,11,5)
(C) (-3,-3,-6) (D) (1,1,2)

61. If $\vec{a}, \vec{b}$ and $\vec{c}$ are non-coplanar vectors and λ is a real number, then the vectors $\vec{a} + 2\vec{b} + 3\vec{c}, \lambda\vec{b} + \mu\vec{c}$ and $(2\lambda - 1)\vec{c}$ are coplanar for

- (A) all values of μ (B) $\lambda = \frac{1}{2}$
(C) $\lambda = 0$ (D) no value of λ

62. The vector $\hat{i} + x\hat{j} + 3\hat{k}$ is rotated through an angle θ and doubled in magnitude and becomes $4\hat{i} + (4x - 2)\hat{j} + 2\hat{k}$. The values of x are

- (A) 1 (B) -2/3 (C) 2 (D) 4/3

63. If unit vectors $\vec{a}$ and $\vec{b}$ are inclined at an angle 2θ such that $|\vec{a} - \vec{b}| < 1$ and $0 \leq \theta \leq \pi$, then θ lies in the interval

- A) $\left[0, \frac{\pi}{6}\right]$ B) $\left(\frac{5\pi}{6}, \pi\right)$
C) $\left[\frac{\pi}{6}, \frac{\pi}{2}\right]$ D) $\left[\frac{\pi}{2}, \frac{5\pi}{6}\right]$

64. If $\vec{a}$ and $\vec{b}$ are non zero vectors such that $|\vec{a} + \vec{b}| = |\vec{a} - 2\vec{b}|$, then

- A) $2\vec{a} \cdot \vec{b} = |\vec{b}|^2$ B) $\vec{a} \cdot \vec{b} = |\vec{b}|^2$
 C) least value of $\vec{a} \cdot \vec{b} + \frac{1}{|\vec{b}|^2 + 2}$ is $\sqrt{2}$
 D) least value of $\vec{a} \cdot \vec{b} + \frac{1}{|\vec{b}|^2 + 2}$ is $\sqrt{2} - 1$

65. The angles of a triangle, two of whose sides are represented by vectors $\sqrt{3}(\hat{a} \times \vec{b})$ and $\hat{b} - (\hat{a} \cdot \vec{b})\hat{a}$, where $\vec{b}$ is a non-zero vector and $\hat{a}$ is a unit vector in the direction of $\vec{b}$, are

- A) $\tan^{-1}(\sqrt{3})$ B) $\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$
 C) $\cot^{-1}(0)$ D) $\tan^{-1}(1)$

66. If $\vec{a} + 2\vec{b} + 3\vec{c} = \vec{0}$, then $\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a} =$

- A) $2(\vec{a} \times \vec{b})$ B) $6(\vec{b} \times \vec{c})$
 C) $3(\vec{c} \times \vec{a})$ D) $\vec{0}$

67. If $\vec{a} \times \vec{b} = \vec{c}$, $\vec{b} \times \vec{c} = \vec{a}$, where $\vec{c} \neq \vec{0}$, then

- A) $|\vec{a}| = |\vec{c}|$ B) $|\vec{a}| = |\vec{b}|$
 C) $|\vec{b}| = 1$ D) $|\vec{a}| = |\vec{b}| = |\vec{c}| = 1$

68. Let $\vec{x}, \vec{y}$ and $\vec{z}$ be three vectors each of magnitude $\sqrt{2}$ and the angle between each pair of them is $\frac{\pi}{3}$. If $\vec{a}$ is a non-zero vector perpendicular to $\vec{x}$ and $\vec{y} \times \vec{z}$ and $\vec{b}$ is a non zero vector perpendicular to $\vec{y}$ and $\vec{z} \times \vec{x}$, then (IIT-2014)

- (A) $\vec{b} = (\vec{b} \cdot \vec{z})(\vec{z} - \vec{x})$ (B) $\vec{a} = (\vec{a} \cdot \vec{y})(\vec{y} - \vec{z})$
 (C) $\vec{a} \cdot \vec{b} = -(\vec{a} \cdot \vec{y})(\vec{b} \cdot \vec{z})$ (D) $\vec{a} = (\vec{a} \cdot \vec{y})(\vec{z} - \vec{y})$

69. If $(x, y, z) \neq (0, 0, 0)$ and $\left(\hat{i} + \hat{j} + 3\hat{k}\right)x + \left(3\hat{i} - 3\hat{j} + \hat{k}\right)y + \left(-4\hat{i} + 5\hat{j}\right)z = a\left(\hat{x}\hat{i} + \hat{y}\hat{j} + \hat{z}\hat{k}\right)$, then the value of a is
 (A) 1 (B) -1 (C) 2 (D) 0

70. A vector $\vec{a}$ has the components $2p$ and 1 w.r.t a rectangular cartesian sytem. This system is rotated through a certain angle about the origin in the counter clockwise sense. If $\vec{a}$ has components $(p+1)$ and 1 with respect to a new system then p is equal to
 (A) -1 (B) -1/3 (C) 1 (D) 2

71. Let $\vec{a}$ and $\vec{b}$ be two non-collinear unit vectors. If $\vec{u} = \vec{a} - (\vec{a} \cdot \vec{b})\vec{b}$ and $\vec{v} = \vec{a} \times \vec{b}$ then $|\vec{v}|$ is [IIT JEE 1999]

- (A) $|\vec{u}|$ (B) $|\vec{u}| + |\vec{u} \cdot \vec{a}|$
 (C) $|\vec{u}| + |\vec{u} \cdot \vec{b}|$ (D) $|\vec{u}| + \vec{u} \cdot (\vec{a} + \vec{b})$

72. If $\vec{a} \times (\vec{b} \times \vec{c})$ is perpendicular to $(\vec{a} \times \vec{b}) \times \vec{c}$ then

- A) $(\vec{a} \cdot \vec{c})|\vec{b}|^2 = (\vec{a} \cdot \vec{b})(\vec{b} \cdot \vec{c})$ B) $\vec{a} \cdot \vec{b} = 0$
 C) $\vec{a} \cdot \vec{c} = 0$ D) $\vec{b} \cdot \vec{c} = 0$

73. Vectors perpendicular to $\hat{i} - \hat{j} - \hat{k}$ and in the plane of $\hat{i} + \hat{j} + \hat{k}$ and $-\hat{i} + \hat{j} + \hat{k}$ are

- A) $\hat{i} + \hat{k}$ B) $2\hat{i} + \hat{j} + \hat{k}$
 C) $3\hat{i} + 2\hat{j} + \hat{k}$ D) $-4\hat{i} - 2\hat{j} - 2\hat{k}$

74. Let $\vec{\alpha} = a\hat{i} + b\hat{j} + c\hat{k}$, $\vec{\beta} = b\hat{i} + c\hat{j} + a\hat{k}$ and $\vec{\gamma} = c\hat{i} + a\hat{j} + b\hat{k}$ be three coplanar vectors with $a \neq b$ and $\vec{v} = \hat{i} + \hat{j} + \hat{k}$. Then $\vec{v}$ is perpendicular to

- A) $\vec{\alpha}$ B) $\vec{\beta}$ C) $\vec{\gamma}$ D) $\vec{\alpha} + \vec{\beta}$

75. If vectors $\vec{A} = 2\hat{i} + 3\hat{j} + 4\hat{k}$, $\vec{B} = \hat{i} + \hat{j} + 5\hat{k}$ and $\vec{C}$ form a left-handed system, then $\vec{C}$ is

VECTOR ALGEBRA

JEE ADVANCED

ASSERTION-REASONING QUESTIONS

- A) $11\hat{i} - 6\hat{j} - \hat{k}$ B) $-11\hat{i} + 6\hat{j} + \hat{k}$
 C) $11\hat{i} - 6\hat{j} + \hat{k}$ D) $-11\hat{i} + 6\hat{j} - \hat{k}$
76. Let $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$, $\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$ and $\vec{c} = \hat{i} + \hat{j} - 2\hat{k}$ three vectors. A vector in the plane of $\vec{b}$ and $\vec{c}$, whose projection on $\vec{a}$ is of magnitude $\sqrt{\frac{2}{3}}$ is [IIT JEE 1993]
 (A) $2\hat{i} + 3\hat{j} - 3\hat{k}$ (B) $2\hat{i} + 3\hat{j} + 3\hat{k}$
 (C) $-2\hat{i} - \hat{j} + 5\hat{k}$ (D) $2\hat{i} + \hat{j} + 5\hat{k}$
77. Which of the following expressions are meaningful [IIT JEE 1998]
 (A) $\vec{u} \cdot (\vec{v} \times \vec{w})$ (B) $(\vec{u} \cdot \vec{v}) \cdot \vec{w}$
 (C) $(\vec{u} \cdot \vec{v}) \vec{w}$ (D) $\vec{u} \times (\vec{v} \cdot \vec{w})$
78. Let $\vec{A}$ be vector parallel to line of intersection of planes P_1 and P_2 through origin. P_1 is parallel to the vectors $2\hat{j} + 3\hat{k}$ and $4\hat{j} - 3\hat{k}$ and P_2 is parallel to $\hat{j} - \hat{k}$ and $3\hat{i} + 3\hat{j}$, then the angle between vector $\vec{A}$ and $2\hat{i} + \hat{j} - 2\hat{k}$ is [IIT JEE 2006]
 (A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{6}$ (D) $\frac{3\pi}{4}$
79. If $\vec{x} \times \vec{b} = \vec{c} \times \vec{b}$ and $\vec{x} \perp \vec{a}$ then $\vec{x}$ is equal to
 (A) $\frac{\vec{b} \times (\vec{a} \times \vec{c})}{\vec{b} \cdot \vec{c}}$ (B) $\frac{(\vec{b} \times \vec{c}) \times \vec{a}}{\vec{b} \cdot \vec{a}}$
 (C) $\frac{\vec{a} \times (\vec{c} \times \vec{b})}{\vec{a} \cdot \vec{b}}$ (D) $\frac{\vec{a} \cdot (\vec{b} \times \vec{c})}{\vec{a} \cdot \vec{b}}$
80. The resolved part of the vector $\vec{a}$ along the vector $\vec{b}$ is $\vec{\lambda}$ and that perpendicular to $\vec{b}$ is $\vec{\mu}$. Then
 (A) $\vec{\lambda} = \frac{(\vec{a} \cdot \vec{b})\vec{a}}{b^2}$ (B) $\vec{\lambda} = \frac{(\vec{a} \cdot \vec{b})\vec{b}}{b^2}$
 (C) $\vec{\mu} = \frac{(\vec{b} \cdot \vec{b})\vec{a} - (\vec{a} \cdot \vec{b})\vec{b}}{b^2}$ (D) $\vec{\mu} = \frac{\vec{b} \times (\vec{a} \times \vec{b})}{b^2}$

- (A) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is a correct explanation for STATEMENT-1
 (B) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is NOT a correct explanation for STATEMENT-1
 (C) STATEMENT-1 is True, STATEMENT-2 is False
 (D) STATEMENT-1 is False, STATEMENT-2 is True
81. Statement 1: If a force $\vec{F}$ passes through $Q(\vec{b})$, then moment of force $\vec{F}$ about $P(\vec{a})$ is $\vec{F} \times \vec{r}$ when $\vec{r} = \overrightarrow{PQ}$
 Statement 2: Moment is a vector
82. Statement 1: If three points P, Q and R have position vectors $\vec{a}, \vec{b}$ and $\vec{c}$ respectively, and $2\vec{a} + 3\vec{b} - 5\vec{c} = 0$ then the points P, Q and R must be collinear
 Statement 2: If for three points A, B and C; $\overrightarrow{AB} = \lambda \overrightarrow{AC}$ then the points A, B and C must be collinear
83. Statement 1: $\vec{u}$ and $\vec{v}$ are unit vectors inclined at an angle α and $\vec{x}$ is a unit vector bisecting the angle between them, then $\vec{x} = (\vec{u} + \vec{v}) / (2 \sin(\alpha/2))$
 Statement 2: If $\triangle ABC$ is an isosceles triangle with $AB = AC = 1$, then the vector representing the bisector of angle A is given by $\overrightarrow{AD} = (\overrightarrow{AB} + \overrightarrow{AC}) / 2$
84. Statement 1: If $\cos \alpha, \cos \beta$ and $\cos \gamma$ are the direction cosines of any line segment, $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$
 Statement 2: If $\cos \alpha, \cos \beta$ and $\cos \gamma$ are the direction cosines of a line segment, $\cos 2\alpha + \cos 2\beta + \cos 2\gamma = -1$
85. Statement 1: The direction cosines of one of the angular bisectors of two intersecting lines having direction cosines as l_1, m_1, n_1 and l_2, m_2, n_2 are proportional to

$$l_1 + l_2, m_1 + m_2, n_1 + n_2$$

Statement 2: The angle between the two intersecting lines having direction cosines as l_1, m_1, n_1 and l_2, m_2, n_2 is given by $\cos \theta = l_1 l_2 + m_1 m_2 + n_1 n_2$

86. **Statement -1 :** $\vec{a} = i + pj + 2k$ and $\vec{b} = 2i + 3j + qk$ are parallel vectors if $p = \frac{3}{2}, q = 4$

Statement-2 : $\vec{a} = a_1 i + a_2 j + a_3 k$ and $\vec{b} = b_1 i + b_2 j + b_3 k$ are parallel $\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3}$

87. **Statement-1 :** If $\vec{a} = 2i + k, \vec{b} = 3j + 4k$ and $\vec{c} = \lambda \vec{a} + \mu \vec{b}$ are coplanar then $\vec{c} = 4\vec{a} - \vec{b}$
Statement-2 : A set vectors $\vec{a}_1, \vec{a}_2, \vec{a}_3, \dots, \vec{a}_n$ is said to be linearly independent if every relation of the form $\ell_1 \vec{a}_1 + \ell_2 \vec{a}_2 + \ell_3 \vec{a}_3 + \dots + \ell_n \vec{a}_n = 0$ implies that $\ell_1 = \ell_2 = \ell_3 = \dots = \ell_n = 0$ (scalar).

88. **Statement -1 :** If I is the incentre of $\triangle ABC$ then $|\overline{BC}| |\overline{IA}| + |\overline{CA}| |\overline{IB}| + |\overline{AB}| |\overline{IC}| = 0$
Statement -2 : The position vector of centroid of $\triangle ABC$ is $\frac{\overline{OA} + \overline{OB} + \overline{OC}}{3}$

PASSAGE TYPE QUESTIONS

- (P) Let the lines BB' and CC' intersect at A. Let $\vec{a}$ be the P.V. of A with respect to origin O. Now, let equations $B'B$ and $C'C$ be respectively $\vec{r} = \vec{a} + t\vec{b}$ and $\vec{r} = \vec{a} + s\vec{c}$. Where t and s are scalars and $\overline{AB}$ is parallel to $\vec{b}$ and $\overline{AC}$ is parallel to $\vec{c}$. Let P be any point on the bisectors of CAB.
 Draw PM perpendicular AB and let $\overline{OP} = \vec{r}$
 $\therefore \angle CAP = \angle PAM = \angle APM$
 $\Rightarrow AM = MP = \lambda$ (say)

$$\therefore \overline{AP} = \overline{AM} + \overline{MP} = \lambda \left(\frac{\vec{b}}{|\vec{b}|} + \frac{\vec{c}}{|\vec{c}|} \right)$$

But $\overline{OP} = \overline{OA} + \overline{AP}$

$$\therefore \vec{r} = \vec{a} + \lambda \left(\frac{\vec{b}}{|\vec{b}|} + \frac{\vec{c}}{|\vec{c}|} \right)$$

Which is the equation of internal bisector of $\angle CAB$.

The bisector AP' of the external $\angle B'AC$ is the bisector of the angle between straight line whose directions are $-\vec{b}$ and $\vec{c}$.

Therefore equation is $\vec{r} = \vec{a} + \mu \left(\frac{\vec{b}}{|\vec{b}|} - \frac{\vec{c}}{|\vec{c}|} \right)$ or

$\vec{r} = \vec{a} + \mu \left(\frac{\vec{c}}{|\vec{c}|} - \frac{\vec{b}}{|\vec{b}|} \right)$ where μ is some scalar.

On the basis of above information, answer the following questions

89. A vector $\vec{d}$, directed along the internal bisector of the angle between the vectors $\vec{b} = 7\hat{i} - 4\hat{j} - 4\hat{k}$ and $\vec{c} = -2\hat{i} - \hat{j} + 2\hat{k}$ and $|\vec{d}| = 5\sqrt{6}$ is
 (A) $\frac{5}{3}(\hat{i} + 7\hat{j} + 2\hat{k})$
 (B) $\frac{5}{3}(5\hat{i} + 5\hat{j} - 2\hat{k})$
 (C) $\frac{5}{3}(\hat{i} - 7\hat{j} + 2\hat{k})$
 (D) $\frac{5}{3}(-5\hat{i} + 5\hat{j} + 2\hat{k})$
90. If the interior and exterior bisectors of the angle A of a triangle ABC meet the base BC at D and E, then

- (A) $\frac{2}{BC} = \frac{1}{BD} + \frac{1}{BE}$
 (B) $2BC = BC + BE$
 (C) $(BC)^2 = BD \times BE$
 (D) $2BD = BC + BE$

91. Let ABC be a triangle and $\vec{a}, \vec{b}, \vec{c}$ be the position vectors of the points A, B, C respectively. External bisectors of $\angle B$ and $\angle C$ meet at P with the sides of the triangle as a, b, c then the position vectors of P becomes

$$(A) -\left(\frac{b\vec{b} + c\vec{c}}{b+c}\right) \quad (B) \left(\frac{a\vec{a} - b\vec{b} - c\vec{c}}{a-b-c}\right)$$

$$(C) abc\left(\frac{\vec{a} + \vec{b} + \vec{c}}{3}\right) \quad (D) \left(\frac{a\vec{a} + b\vec{b} + c\vec{c}}{a+b+c}\right)$$

- (P) Let O, N, G and O' are the circumcentre, centre of nine point circle, centroid and orthocenter of a triangle ABC respectively. AL and BM are perpendiculars from A and B on sides BC and CA respectively. Let AD be the median and OD is perpendicular to side BC. Let R be the circum radius of triangle ABC, then $OA = OB = OC = R$.

Now, in $\triangle OBD$, $OD = R \cos A$, in $\triangle ABM$,

$$AO' = AM \sec(90^\circ - C) (\because \angle O'AM = 90^\circ - C)$$

$$= AM \cos \sec C = \frac{C \cos A}{\sin C} = 2R \cos A$$

$$\therefore AO' = 2OD$$

If S be any point in the plane of triangle ABC and AP is the diameter of the circumcircle.

On the basis of above information, answer the following questions

92. $\vec{AO'} + \vec{O'B} + \vec{O'C}$ is equal to

$$(A) \vec{OO'} \quad (B) 2\vec{OO'} \quad (C) 2\vec{AO} \quad (D) 3\vec{SG}$$

93. $\vec{NA} + \vec{NB} + \vec{NC}$ is equal to

$$(A) \frac{1}{2}\vec{OO'} \quad (B) \vec{OO'} \quad (C) 2\vec{O'O} \quad (D) 2\vec{NG}$$

94. $\vec{SA} + \vec{SB} + \vec{SC}$ is equal to

$$(A) \vec{SG} \quad (B) 2\vec{SG} \quad (C) 3\vec{SG} \quad (D) \frac{1}{2}\vec{SG}$$

- (P) Let O be the origin, $\vec{OA} = \vec{a}$, $\vec{OB} = \vec{b}$ be two vectors. Then the point $\vec{OC} = p\vec{a} + q\vec{b}$ lies

(i) inside $\triangle OAB$, if $p > 0$, $q > 0$ and $p + q < 1$.

(ii) outside $\triangle OAB$ but inside $\angle AOB$ if $p > 0$, $q > 0$ and $p + q > 1$.

(iii) outside $\triangle OAB$ but inside $\angle OAB$ if $p < 0$, $q > 0$ and $p + q < 1$.

(iv) outside $\triangle OBA$ but inside $\angle OBA$ if $p > 0$, $q < 0$ and $p + q < 1$.

On the basis of above information answer the following

95. $\vec{c} = \frac{1}{2}\vec{a} + \frac{1}{3}\vec{b}$

- (A) inside $\triangle OAB$,
(B) outside $\triangle OAB$ but inside $\angle AOB$
(C) outside $\triangle OAB$ but inside $\angle OAB$
(D) outside $\triangle OBA$ but inside $\angle OBA$

96. $\frac{1}{2}\vec{c} = \frac{1}{2}\vec{a} + \frac{2}{3}\vec{b}$

- (A) inside $\triangle OAB$,
(B) outside $\triangle OAB$ but inside $\angle AOB$
(C) outside $\triangle OAB$ but inside $\angle OAB$
(D) outside $\triangle OBA$ but inside $\angle OBA$

97. $\vec{c} = -\frac{1}{2}\vec{a} + \frac{3}{4}\vec{b}$

- (A) inside $\triangle OAB$
(B) outside $\triangle OAB$ but inside $\angle AOB$
(C) outside $\triangle OAB$ but inside $\angle OAB$
(D) outside $\triangle OBA$ but inside $\angle OBA$

98. $\vec{c} = \frac{3}{4}\vec{a} - \frac{1}{2}\vec{b}$

- (A) inside $\triangle OAB$
(B) outside $\triangle OAB$ but inside $\angle AOB$
(C) outside $\triangle OAB$ but inside $\angle OAB$
(D) outside $\triangle OBA$ but inside $\angle OBA$

- (P) Consider a line $\vec{r} = \vec{i} + t(\hat{i} + \hat{j} + \hat{k})$ and a plane $(\vec{r} - (\hat{i} + \hat{j})) \cdot (\hat{i} - \hat{j} - \hat{k}) = 1$

99. The angle between the line and the plane is

- (A) $\cos^{-1} 1/3$ (B) $\sin^{-1} 1/3$
(C) $\pi/4$ (D) $\pi/2$

100. The position vector of the point of intersection of the line and the plane is

- (A) $\hat{i}$ (B) $\hat{j}$ (C) $\hat{k}$ (D) $\hat{i} + \hat{j}$

101. The position vector of the point on the given line whose distance from the plane is

$$\frac{1}{\sqrt{3}} \text{ units is}$$

(A) $-\hat{i} - 2\hat{j} - 2\hat{k}$ (B) $\hat{i}$

(C) $2\hat{i} + \hat{j} + \hat{k}$ (D) $3\hat{i} + 2\hat{j} + 2\hat{k}$

- (P) A new operation $*$ is defined between two non parallel vectors $\vec{\alpha}$ and $\vec{\beta}$ as

$\vec{\alpha} * \vec{\beta} = |\vec{\alpha}| |\vec{\beta}| \tan\left(\frac{\theta}{2}\right)$, where θ is the angle between $\vec{\alpha}$ and $\vec{\beta}$.

102. The condition for which $\vec{\alpha}$ and $\vec{\beta}$ are perpendicular is

(A) $\vec{\alpha} * \vec{\beta} = 0$ (B) $\frac{\vec{\alpha} * \vec{\beta}}{|\vec{\alpha}| |\vec{\beta}|} = 1$

(C) $\frac{\vec{\alpha} * \vec{\beta}}{|\vec{\alpha}| |\vec{\beta}|} = -1$ (D) $\frac{\vec{\alpha} * \vec{\beta}}{|\vec{\alpha}|} = 1$

103. $\vec{\alpha} * \vec{\alpha}$ is

(A) $|\vec{\alpha}|^2$ (B) not defined

(C) 0 (D) $|\vec{\alpha}|^2 - 3$

104. For $\vec{\alpha} * \vec{\beta} = \vec{\alpha} \cdot \vec{\beta}$

(A) $|\vec{\alpha}| = 0$ is a necessary condition

(B) $|\vec{\alpha}| \cdot |\vec{\beta}| = 0$ is a necessary condition

(C) $t^3 + t^2 + t = 1$ is a sufficient condition, where $t = \tan \frac{\theta}{2}$ (D) $|\vec{\alpha}| = |\vec{\beta}|$ is a necessary condition

- (P) Let ABC be a triangle, AD, BE, CF be the angular bisectors of its interior angles. These bisectors are concurrent at a point I called incentre of the triangle. we know that

from geometry that $\frac{BD}{DC} = \frac{AB}{AC}$. If $BC = \alpha$,

$CA = \beta$ and $AB = \gamma$ and with reference to the same origin. Let $\vec{a}, \vec{b}, \vec{c}$ be position vectors of A, B, C respectively, then

105. The position vector of I must be

(A) $\frac{\vec{a} + \vec{b} + \vec{c}}{3}$ (B) $\frac{\alpha \vec{a} + \beta \vec{b} + \gamma \vec{c}}{3}$

(C) $\frac{\alpha \vec{a} + \beta \vec{b} + \gamma \vec{c}}{\alpha + \beta + \gamma}$ (D) $\frac{\alpha \vec{a} + \beta \vec{b} + \gamma \vec{c}}{2}$

106. If 'r' is the perpendicular distance of I from the side BC, then $\vec{IB} \cdot \vec{IC}$ must be

a) $r^2 \cos ec\left(\frac{B}{2}\right) \cos ec\left(\frac{C}{2}\right)$

b) $r^2 \cos ec\left(\frac{B}{2}\right) \cos ec\left(\frac{C}{2}\right) \sin\left(\frac{A}{2}\right)$

c) $-r^2 \cos ec\left(\frac{B}{2}\right) \cos ec\left(\frac{C}{2}\right) \sin\left(\frac{A}{2}\right)$

d) $-r^2 \cos ec\left(\frac{A}{2}\right) \cos ec\left(\frac{B}{2}\right) \sin\left(\frac{C}{2}\right)$

107. If 'r' is the perpendicular distance of I from the side BC, then $|\vec{IB} \times \vec{IC}|$ must be

a) $r^2 \cos ec\left(\frac{B}{2}\right) \cos ec\left(\frac{C}{2}\right)$

b) $r^2 \cos ec\left(\frac{B}{2}\right) \cos ec\left(\frac{C}{2}\right) \cos\left(\frac{A}{2}\right)$

c) $r^2 \cos ec\left(\frac{B}{2}\right) \cos ec\left(\frac{C}{2}\right) \cos ec\left(\frac{A}{2}\right)$

d) $r^2 \cos ec\left(\frac{B}{2}\right) \cos ec\left(\frac{C}{2}\right) \sin\left(\frac{A}{2}\right)$

- (P) ABCD is a tetrahedron having each edge of unit length.

108. The volume of the tetrahedron is

a) $\frac{1}{\sqrt{2}}$ b) $\frac{1}{2\sqrt{2}}$ c) $\frac{1}{3\sqrt{2}}$ d) $\frac{1}{6\sqrt{2}}$

109. The angle between two adjacent faces is

a) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$ b) $\cos^{-1}\left(\frac{1}{3}\right)$

c) $\frac{\pi}{3}$ d) $\frac{\pi}{4}$

110. The distance between a pair of opposite edges is

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- a) 1 b) $\frac{1}{2}$ c) $\frac{1}{\sqrt{2}}$ d) $\frac{1}{\sqrt{3}}$

(P) P is a point on the straight line $\vec{r} = 5\hat{i} + 7\hat{j} - 2\hat{k} + s(3\hat{i} - \hat{j} + \hat{k})$ and Q is a point on the straight line $\vec{r} = -3\hat{i} + 3\hat{j} + 6\hat{k} + t(-3\hat{i} + 2\hat{j} + 4\hat{k})$ such that PQ is parallel to the vector $2\hat{i} + 7\hat{j} - 5\hat{k}$.

111. The position vector of P is

- (a) $\hat{i} + 7\hat{j} - 4\hat{k}$ (b) $2\hat{i} + 6\hat{j} - 5\hat{k}$
(c) $\hat{i} + \hat{j} - \hat{k}$ (d) $2\hat{i} + 8\hat{j} - 3\hat{k}$

112. The position vector of Q is

- (a) $3\hat{i} + 14\hat{j} - 9\hat{k}$ (b) $\hat{j} + 2\hat{k}$
(c) $-\hat{i} - 6\hat{j} + 4\hat{k}$ (d) $-\hat{j}$

113. The distance PQ is

- (a) $\sqrt{78}$ (b) $\sqrt{71}$ (c) 8 (d) $\sqrt{50}$

(P) The vertices of a $\triangle ABC$ are A(3, 1, 3),

B(0, 2, 2) and C is (2, -1, 5). Points D and E

divide AB, CA in the ratio 1 : 2 respectively.

P is a point on space such that perpendicular from it to the plane of ABC meets at F,

where F is the point of intersection of CD

and BE. Also the distance of P from the plane

of $\triangle ABC$ is $\sqrt{2}$

114. $\overrightarrow{FP}$ is

- (A) $7\hat{i} + 7\hat{j} + 7\hat{k}$ (B) $\frac{7}{2}(\hat{j} + \hat{k})$
(C) $7\hat{j} + 7\hat{k}$ (D) $\hat{j} + \hat{k}$

115. The volume of tetrahedron PABC is

- (A) $\frac{4}{7}$ cubic units (B) 7 cubic units

- (C) $\frac{7}{3}$ cubic units (D) $\frac{7}{4}$ cubic units

116. The vector equation of line AP is $\vec{r} =$

(where 't' being a parameter)

- A) $3\hat{i} + \hat{j} + 3\hat{k} + t(\hat{i} + \hat{j} + \hat{k})$
B) $3\hat{i} + \hat{j} + 3\hat{k} + t(-\hat{i} - \hat{j} + \hat{k})$
C) $3\hat{i} + \hat{j} + 3\hat{k} + t(-\hat{i} - \hat{j} - \hat{k})$
D) $3\hat{i} + \hat{j} + 3\hat{k} + t(\hat{i} - 2\hat{k})$

(P) Let $\vec{r}$ is a position vector of a variable point

in Cartesian OXY plane such that

$$\vec{r} \cdot (10\hat{j} - 8\hat{i} - \vec{r}) = 40 \text{ and}$$

$$p_1 = \max \left\{ \left| \vec{r} + 2\hat{i} - 3\hat{j} \right|^2 \right\},$$

$$p_2 = \min \left\{ \left| \vec{r} + 2\hat{i} - 3\hat{j} \right|^2 \right\} \text{ then}$$

117. p_2 is equal to

- (A) 9 (B) $2\sqrt{2} - 1$
(C) $6\sqrt{2} + 3$ (D) $9 - 4\sqrt{2}$

118. $p_1 + p_2$ is equal to

- (A) $4\sqrt{2}$ (B) 10 (C) 18 (D) $5\sqrt{2}$

119. $(p_1 p_2)^{1/2} =$

- (A) 1 (B) 2 (C) $\sqrt{7}$ (D) 4

MATRIX-MATCHING QUESTIONS

120. The diagonals of a parallelogram ABCD intersect at E. O is the origin of reference. If position vector of A is a, $\overrightarrow{AB} = \vec{p}$ and $\overrightarrow{BC} = \vec{q}$.

Column I

Column II

- | | |
|---------------------------------------|--------------------------------------|
| (A) Position vector of midpoint of CD | (p) $(1/2)(2a + \vec{p} + \vec{q})$ |
| (B) Position vector of E | (q) $a + \vec{p} + \vec{q}$ |
| (C) Position vector of C | (r) $(1/2)(2a + \vec{p} + 2\vec{q})$ |
| (D) Position vector of D | (s) $a + \vec{q}$ |

121. If D, E, F are mid points of the sides BC, CA, AB respectively of a triangle ABC and O is any point then

Column I(A) $AD + BE + CF$ is equal(B) $OA + OB + OC$ (C) $OE + OF + DO$ (D) $AD + \left(\frac{2}{3}\right)BE + \left(\frac{1}{3}\right)CF$ **Column II**(p) $\left(\frac{1}{2}\right)AC$ (q) OA (r) $OD + OE + OF$ (s) O **122. Column I**

A) If $|\vec{a}| = |\vec{b}| = |\vec{c}|$, angle between each pair of vectors is $\frac{\pi}{3}$ and $|\vec{a} + \vec{b} + \vec{c}| = \sqrt{6}$, then $2|\vec{a}|$ is equal to

B) If $\vec{a}$ is a perpendicular to $\vec{b} + \vec{c}$, $\vec{b}$ is perpendicular to $\vec{c} + \vec{a}$, $\vec{c}$ is perpendicular to $\vec{a} + \vec{b}$, $|\vec{a}| = 2, |\vec{b}| = 3$ and $|\vec{c}| = 6$. Then $|\vec{a} + \vec{b} + \vec{c}| - 2$ is equal to

C) $\vec{a} = 2\hat{i} + 3\hat{j} - \hat{k}$, $\vec{b} = -\hat{i} + 2\hat{j} - 4\hat{k}$, $\vec{c} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{d} = 3\hat{i} + 2\hat{j} + \hat{k}$ then $\frac{1}{7}(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d})$ is equal to

D) If $|\vec{a}| = |\vec{b}| = |\vec{c}| = 2$ and $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = 2$, then $[\vec{a} \vec{b} \vec{c}] \cos 45^\circ$ is equal to

Column II

p. 3

q. 2

r. 4

s. 5

123. Matrix match type**Column - I**

(A) Let $|\vec{a}| = 1, |\vec{b}|^2 = 2, |\vec{c}|^2 = 3$ and

$\vec{a} \perp (\vec{b} + \vec{c}), \vec{b} \perp (\vec{c} + \vec{a}), \vec{c} \perp (\vec{a} + \vec{b})$ then $|\vec{a} + \vec{b} + \vec{c}|$ is

(B) If $\vec{a}, \vec{b}, \vec{c}$ are three vectors of equal magnitude and the angle between each pair of vectors is $\frac{\pi}{3}$

such that $|\vec{a} + \vec{b} + \vec{c}| = \sqrt{6}$ then $|\vec{a}|$ is

(C) If $|\vec{a}| = 5, |\vec{a} - \vec{b}| = 8$ and $|\vec{a} + \vec{b}| = 10$ then $|\vec{b}|$ is

(D) If $\vec{a}, \vec{b}, \vec{c}$ are three non zero vectors such that $\vec{a} + \vec{b} + \vec{c} = 0$ then the value of $\vec{a} \cdot (\vec{b} + \vec{a}) + \vec{b} \cdot (\vec{c} + \vec{b}) + \vec{c} \cdot (\vec{a} + \vec{c})$ may be

Column - II

(p) 1

(q) $\sqrt{57}$ (r) $\sqrt{6}$

(s) 0

(t) 8

124. Match list with list II and select the correct answer using the code given below the lists (IIT - 2013)**List - I**

P) Volume of parallelopiped determined by vectors $\vec{a}, \vec{b}$ and $\vec{c}$ is 2. Then the volume of the parallelopiped determined by vectors $2(\vec{a} \times \vec{b}), 3(\vec{b} \times \vec{c})$ and $(\vec{c} \times \vec{a})$

Q) Volume of parallelopiped determined by vectors $\vec{a}, \vec{b}$ and $\vec{c}$ is 5. Then the volume of the parallelopiped determined by vectors $3(\vec{a} + \vec{b}), (\vec{b} + \vec{c})$ and $2(\vec{c} + \vec{a})$ is

R) Area of a triangle with adjacent sides determined by vectors $\vec{a}$ and $\vec{b}$ is 20. Then the area of the triangle with adjacent sides determined by vectors $(2\vec{a} + 3\vec{b})$ and $(\vec{a} - \vec{b})$ is

S) Area of a parallelogram with adjacent sides determined by vectors $\vec{a}$ and $\vec{b}$ is 30. Then the area of the parallelogram with adjacent sides determined by vectors $(\vec{a} + \vec{b})$ and $\vec{a}$ is

List - II

1) 100

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- 2) 30
3) 24
4) 60

	P	Q	R	S
A)	4	2	3	1
B)	2	3	1	4
C)	3	4	1	2
D)	1	4	3	2

125. Column - I

- (A) For given vectors $\vec{a}(2, 1, -1), \vec{b}(1, 2, 1), \vec{c}(2, -1, 3)$ and $\vec{d}(3, -1, 2)$ the projection of the vector $\vec{a} + \vec{c}$ on the vector $(\vec{b} - \vec{d}) \times \vec{c}$ is
- (B) If $\vec{a}, \vec{b}$ and $\vec{c}$ are unit vectors satisfying $\vec{a} + \vec{b} + \vec{c} = 0$ and $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}$ is equal to
- (C) If $(\vec{a} \times \vec{b}) \cdot (\vec{b} \times \vec{c}) \times (\vec{c} \times \vec{a}) = [\vec{a} \cdot (\vec{b} \times \vec{c})]^k$, then k is
- (D) The value of c for which the vectors $\vec{p} = (c \log_2 x, -6, 3)$ and $\vec{q} = (\log_2 x, 2, 2c \log_2 x)$ make an obtuse angle for any $x \in (0, \infty)$ are

Column - II

- (p) $\left(-\frac{4}{3}, 0\right)$
(q) $\sqrt{6}$
(r) $-\frac{3}{2}$
(s) 2
(t) -1

126. Column I

- (A) The line $\vec{r} = (\hat{i} + \hat{j}) + t(\hat{i} - \hat{k})$ where t is scalar passes through the point
- (B) The line $\vec{r} = (\hat{i} + \hat{j}) + t(\hat{i} - \hat{k})$ where t is scalar and the plane $\vec{r} \cdot (2\hat{i} + \hat{j} + \hat{k}) = 2$ intersect at the point
- (C) The point on the line $\vec{r} = (\hat{i} + \hat{j}) + t(\hat{i} - \hat{k})$ where t is scalar, which is at a distance of 3 units from the point having position vector $\hat{i}$ is/are
- (D) The volume of the parallelopiped having

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adjacent sides $\hat{i} + \hat{k}, 2\hat{i} + \hat{j} + \hat{k}$ and $\vec{C}$ is 4 cubic units then $\vec{C}$ may be

Column II

- (p) $-\hat{i} - \hat{j} + 2\hat{k}$
(q) $-\hat{i} + \hat{j} + 2\hat{k}$
(r) $-2\hat{i} + \hat{j} + 3\hat{k}$
(s) $\hat{j} + \hat{k}$
(t) $3\hat{i} + \hat{j} - 2\hat{k}$

127. Matrix match type

Column-I

- (A) The angle between the lines given by $\vec{r} = 3\hat{i} + 2\hat{j} - 4\hat{k} + \lambda(\hat{i} + 2\hat{j} + 2\hat{k})$ and $\vec{r} = 5\hat{i} - 2\hat{j} + \mu(3\hat{i} + 2\hat{j} + 6\hat{k})$ is
- (B) A vector perpendicular to the plane containing the vectors $\hat{i} - 2\hat{j} - \hat{k}$ and $3\hat{i} - 2\hat{j} - \hat{k}$ is inclined to the vector $\hat{i} + \hat{j} + \hat{k}$ at an angle
- (C) In a cube the angle between face diagonal and body diagonal is
- (D) If $\vec{a}, \vec{b}, \vec{u}, \vec{v}, \vec{w}, \vec{z}$ are non zero vectors such that $[\vec{u}, \vec{v}, \vec{w}]\vec{a} = [\vec{u}, \vec{v}, \vec{z}]\vec{b}$ where $\vec{a}$ and $\vec{b}$ are linearly independent then the angle between $\vec{v} \times \vec{w}$ and $2\vec{u} + 3\vec{z}$ is

Column - II

- (p) 0
(q) $\cos^{-1}\left(\frac{1}{\sqrt{15}}\right)$
(r) $\cos^{-1}\left(\frac{\sqrt{2}}{\sqrt{3}}\right)$
(s) $\cos^{-1}\left(\frac{19}{21}\right)$
(t) $\frac{\pi}{2}$

INTEGER QUESTIONS

128. ABCD is a parallelogram A_1 and B_1 are midpoints of sides BC and CD respectively.

If $\overline{AA_1} + \overline{AB_1} = \lambda \overline{AC}$ then 2λ

129. Let 'G' be the centroid of the triangle ABC

whose sides are of length a, b, c if 'P' is point in the plane of triangle ABC, such that $PA=1$,

$PB=3$, $PC=2$, $PG=2$ then $(a^2 + b^2 + c^2) =$

130. Let $\vec{A}$ and $\vec{B}$ be two non-parallel unit vectors in a plane. If $(\alpha \vec{A} + \vec{B})$ bisects the internal angle between $\vec{A}$ and $\vec{B}$, then the value of α is

131. Let $|\vec{p}| = \frac{2}{3}\sqrt{2}$, $|\vec{q}| = 1$ and the angle between $\vec{p}$ and $\vec{q}$ be $\frac{\pi}{4}$. If a parallelogram is formed with adjacent sides $\vec{a} = \vec{p} - 3\vec{q}$ and $\vec{b} = 5\vec{p} + 2\vec{q}$, then the length of the shorter diagonal is

132. Two points P and Q are given in the rectangular Cartesian co-ordinates on the curve $y = 2^{x+2}$, such that $\vec{OP} \cdot \vec{i} = -1$, $\vec{OQ} \cdot \vec{i} = -2$ where $\vec{i}$ is a unit vector along the x-axis. then $\frac{1}{\sqrt{5}}(4|\vec{OQ}| - |\vec{OP}|) =$

133. Let PQ and QR be diagonals of adjacent faces of a rectangular box, with its center at O. If $\angle QOR, \angle ROP, \angle POQ$ are θ, ϕ, ψ respectively, then the value of $|\cos \theta + \cos \phi + \cos \psi| =$

134. If $\vec{a} = x\hat{i} + (x-1)\hat{j} + \hat{k}$ and $\vec{b} = (x+1)\hat{i} + \hat{j} + x\hat{k}$ always make an acute angle for all $x \in R$, then the least integral value of x is

135. Let $\vec{u} = \hat{i} + \hat{j}, \vec{v} = \hat{i} - \hat{j}$, and $\vec{w} = \hat{i} + 2\hat{j} + 3\hat{k}$. If $\hat{n}$ is a unit vector such that $\vec{u} \cdot \hat{n} = 0$ and $\vec{v} \cdot \hat{n} = 0$, then the value of $|\vec{w} \cdot \hat{n}|$ is

136. If $\vec{a}, \vec{b}$ and $\vec{c}$ are unit vectors satisfying $|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2 = 9$, then

$$|2\vec{a} + 5\vec{b} + 5\vec{c}| \text{ is } \quad [\text{IITJEE 2012}]$$

137. Given $|\vec{a}| = |\vec{b}| = 1$ and $|\vec{a} + \vec{b}| = \sqrt{3}$. If $\vec{c}$ is a vector such that $\vec{c} - \vec{a} - 2\vec{b} = 3(\vec{a} \times \vec{b})$, then the value of $[\vec{c}, \vec{b}]$, where $[\cdot]$ represents the greatest integer function, is

138. Let O be an interior point of $\triangle ABC$ such that $\vec{OA} + 2\vec{OB} + 3\vec{OC} = \vec{0}$. Then the ratio of the area of $\triangle ABC$ to the area of $\triangle AOC$ is

139. Let $A_1, A_2, \dots, A_n$ ($n > 2$) be the vertices of a regular polygon of 'n' sides with its centre at the origin. Let $\vec{a}_k$ be the position vector of the point $A_k, k = 1, 2, \dots, n$. If

$$\left| \sum_{k=1}^{n-1} (\vec{a}_k \times \vec{a}_{k+1}) \right| = \left| \sum_{k=1}^{n-1} (\vec{a}_k \cdot \vec{a}_{k+1}) \right| \text{ then the}$$

maximum value of 'n' is (IIT JEE-2014)

140. The position vectors of the points A, B, C, D are $3\hat{i} - 2\hat{j} - \hat{k}$, $2\hat{i} + 3\hat{j} - 4\hat{k}$, $-\hat{i} + \hat{j} + 2\hat{k}$, $4\hat{i} + 5\hat{j} + \lambda\hat{k}$. If the points A, B, C, D are coplanar then $\frac{-17\lambda}{73} =$

141. If $\vec{a}, \vec{b}, \vec{c}$ are three vectors such that $[\vec{a} \vec{b} \vec{c}] = 1$ then the value of $[\vec{a} + \vec{b} \vec{b} + \vec{c} \vec{c} + \vec{a}] + [\vec{a} \times \vec{b} \vec{b} \times \vec{c} \vec{c} \times \vec{a}] + [\vec{a} \times (\vec{b} \times \vec{c}), \vec{b} \times (\vec{c} \times \vec{a}), \vec{c} \times (\vec{a} \times \vec{b})]$ is

142. The three vectors $\hat{i} + \hat{j}, \hat{j} + \hat{k}, \hat{k} + \hat{i}$ taken two at a time form three planes. If V be the volume of the tetrahedron having adjacent sides as the three unit vectors drawn perpendicular to those three planes then the value of $9\sqrt{3}V$ is

143. Let $\ell(\vec{b} \times \vec{c}) + m(\vec{c} \times \vec{a}) + n(\vec{a} \times \vec{b}) = 4\hat{i}$, $[\vec{a} \vec{b} \vec{c}] = 2$ and $\vec{a} + \vec{b} + \vec{c} = 3\hat{i}$ then the value of $(\ell + m + n)$ is

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144. If h is the altitude of a parallelepiped determined by the vectors $\vec{a}, \vec{b}, \vec{c}$ and the base is taken to be the parallelogram determined by $\vec{a}$ and $\vec{b}$ where $\vec{a} = \hat{i} + \hat{j} + \hat{k}, \vec{b} = 2\hat{i} + 4\hat{j} - \hat{k}$ and $\vec{c} = \hat{i} + \hat{j} + 3\hat{k}$ then the value of $19h^2$ is
145. Let F be the foot of perpendicular from the point with position vector $\hat{i} + 3\hat{j} + 4\hat{k}$ on the plane $\vec{r} \cdot (2\hat{i} - \hat{j} + \hat{k}) + 3 = 0$. The distance of F from the x - axis is
146. Let P be the image of the point with position vector $2\hat{i} - 3\hat{j} + 3\hat{k}$ in the plane $\vec{r} \cdot (\hat{i} - 2\hat{j} - \hat{k}) + 1 = 0$. The distance of P from z- axis is
147. If $\vec{a}, \vec{b}$ and $\vec{c}$ are vectors such that $|\vec{a}| = 3, |\vec{b}| = 4$, and $|\vec{c}| = 5$ and $(\vec{a} + \vec{b})$ is perpendicular to $\vec{c}, (\vec{b} + \vec{c})$ is perpendicular to $\vec{a}$ and $(\vec{c} + \vec{a})$ is perpendicular to $\vec{b}$ then the value of $\left[\vec{a} + \vec{b} + \vec{c} \right]$ (where $[.]$ represents the greatest integer function) is
148. Let G_1, G_2 and G_3 be the centroids of the triangular faces OBC, OCA, and OAB of a tetrahedron OABC. If V_1 denotes the volume of the tetrahedron OABC and V_2 that of the parallelepiped with OG_1, OG_2 and OG_3 as three concurrent edges then the value of $4V_1/V_2$ is
149. If $\vec{e}_1, \vec{e}_2, \vec{e}_3$ and $\vec{E}_1, \vec{E}_2, \vec{E}_3$ are two set of vectors such that $\vec{e}_i \cdot \vec{E}_j = 1$, if $i = j$ and $\vec{e}_i \cdot \vec{E}_j = 0$, if $i \neq j$ then the value of $[\vec{e}_1, \vec{e}_2, \vec{e}_3][\vec{E}_1, \vec{E}_2, \vec{E}_3]$ is
150. Let $\vec{a} = 2\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{b} = \hat{i} + \hat{j}$. If $\vec{c}$ is a vector such that $\vec{a} \cdot \vec{c} = |\vec{c}|, |\vec{c} - \vec{a}| = 2\sqrt{2}$ and the angle between $\vec{a} \times \vec{b}$ and $\vec{c}$ is 30° then the value of $\left| 2(\vec{a} \times \vec{b}) \times \vec{c} \right|$ is

151. Let $\vec{V} = 2\vec{i} + \vec{j} - \vec{k}$ and $\vec{W} = \vec{i} + 3\vec{k}$. If $\vec{U}$ is a unit vector, then the maximum integral value of the scalar triple product $[\vec{U}\vec{V}\vec{W}]$ is
152. Let $\vec{PR} = 3\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{SQ} = \hat{i} - 3\hat{j} - 4\hat{k}$ determine diagonals of a parallelogram PQRS and $\vec{PT} = \hat{i} + 2\hat{j} + 3\hat{k}$ be another vector. If the volume of the parallelepiped determined by the vector $\vec{PT}, \vec{PQ}$ and $\vec{PS}$ is k then $k - 1$ is (IIT - 2013)
153. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three non-coplanar unit vectors such that the angle between every pair of them is $\frac{\pi}{3}$ if $\vec{a} \times \vec{b} + \vec{b} \times \vec{c} = p\vec{a} + q\vec{b} + r\vec{c}$ where p, q and r are scalars, then the value of $\frac{p^2 + 2q^2 + r^2}{q^2} =$ (IIT-JEE-2014)
154. If G is the centroid of $\triangle ABC$ having lengths of the sides a, b, c and 'O' is any point then $a^2 + b^2 + c^2 = 3(\vec{OA}^2 + \vec{OB}^2 + \vec{OC}^2) - k(\vec{OG})^2$ where k = (IIT-JEE-2014)

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KEY**SINGLE ANSWER QUESTIONS**

1.D	2.B	3.A	4.B
5.C	6.A	7.D	8.B
9.C	10.A	11.A	12.B
13.A	14.C	15.C	16.A
17.C	18.D	19.C	20.C
21.A	22.A	23.B	24.D
25.C	26.A	27.B	28.B
29.B	30.C	31.D	32.D
33.A	34.A	35.C	36.D
37.B	38.D	39.C	40.B
41.A	42.C	43.B	44.C
45.C	46.C	47.A	48.A
49.B	50.B	51.A	52.B
53.A	54.A		

MULTI ANSWER QUESTIONS

55.AD	56.AC	57.AC	58.ABCD
59.AC	60.ACD	61.ABC	62.BC
63.AB	64.AD	65.ABC	66.ABC
67.AC	68.ABC	69.BD	70.BC
71.AC	72.AC	73.BD	74.ABCD
75.BD	76.AC	77.AC	78.BD
79.BC	80.BCD		

ASSERTION-REASONING QUESTIONS

81. D	82. A	83. D	84. B
85. B	86. A	87. B	88. B

PASSAGE QUESTIONS

89. C	90. A	91. B	92. A
93. A	94. C	95. A	96. B
97. C	98. D	99. B	100. A
101. C	102. B	103. C	104. C
105. C	106. C	107. B	108. D
109. B	110. C	111. D	112. B
113. A	114. D	115. C	116. D
117. B	118. A	119. C	

MATRIX - MATCHING QUESTIONS

120. (A) - r, (B) - p, (C) - q, (D) - s
 121. (A) - s, (B) - r, (C) - q, (D) - p

122. $a \rightarrow q; b \rightarrow s; c \rightarrow p; d \rightarrow r$

123. (A)-(r); (B)-(p); (C)-(q);
 (D)-(p,q,r,t)

124. $P-3, Q-4, R-1, S-2$

125. (A)-(q); (B)-(r); (C)-(s); (D)-(p,t)

126. (A)-(q,r,s,t); (B)-(s); (C)-(q,t);
 (D)-(q,t)

127. A-q; B-q; C-r; D-t

INTEGER QUESTIONS

128.3	129.6	130.1	131.5
132.3	133.1	134.3	135.3
136.3	137.2	138.3	139.8
140.2	141.3	142.2	143.6
144.8	145.5	146.1	147.7
148.9	149.1	150.3	151.7
152.9	153.4	154.9	

HINTS**SINGLE ANSWER QUESTIONS**

1. $\overline{OE} = \frac{\overline{OA} + \overline{OB} + \overline{OC} + \overline{OD}}{4}$

2. Let the incentre be at the origin and be $A(\vec{p})$, $B(\vec{q})$ and $C(\vec{r})$. Then $\overline{IA} = \vec{p}$, $\overline{IB} = \vec{q}$, $\overline{IC} = \vec{r}$.

Incentre is $\frac{a\vec{p} + b\vec{q} + c\vec{r}}{a + b + c}$, where $p = BC$, $q =$

AC and $r = AB$.

Incentre is at the origin. Therefore,

$$\frac{a\vec{p} + b\vec{q} + c\vec{r}}{a + b + c} = \vec{0}, \text{ or } a\vec{p} + b\vec{q} + c\vec{r} = \vec{0}$$

$$\Rightarrow a\overline{IA} + b\overline{IB} + c\overline{IC} = \vec{0}$$

3. $\overline{AD} = \frac{\overline{AB} + \overline{AC}}{2}, \overline{BE} = \frac{\overline{BA} + \overline{BC}}{2}, \overline{CF} = \frac{\overline{CA} + \overline{CB}}{2}$

4. $\vec{p} = \lambda \vec{q}$

$$(x+4)\vec{a} + \vec{b} = \lambda[(2-2x)\vec{b} + 2\vec{b}]$$

VECTOR ALGEBRA

JEE ADVANCED

$$\Rightarrow \lambda = \frac{1}{2}$$

$$2x = -3 \Rightarrow x = -\frac{3}{2}$$

$$\begin{aligned} 5. \quad \text{Sum} &= (\overline{AB} + \overline{BC} + \dots + \overline{EA}) + \\ &(\overline{BC} + \dots + \overline{EA}) + (\overline{CD} + \dots + \overline{EA}) + (\overline{DE} + \overline{EA}) + \overline{EA} \\ &= \overline{AA} + \overline{BA} + \overline{CA} + \overline{DA} + \overline{EA} \\ &= 4\overline{OA} - (\overline{OB} + \overline{OC} + \overline{OD} + \overline{OE}) \\ &\because \overline{OA} + \overline{OB} + \overline{OC} + \overline{OD} + \overline{OE} = \overline{O} \\ &= 5\overline{OA} = 5\vec{a} \end{aligned}$$

$$6. \quad \overline{AB} + \overline{BC} = \overline{AC} \Rightarrow \overline{BC} = \frac{\vec{u}}{|\vec{u}|} + \frac{\vec{v}}{|\vec{v}|} \text{ and}$$

$$\overline{AB} \cdot \overline{BC} = 0 \Rightarrow 1 + \cos 2A + \cos 2B + \cos 2C = 0$$

$$7. \quad \text{Vector along angular bisector is } \lambda(\vec{a} + \vec{b})$$

$$\therefore \text{Unit vector} = \frac{\vec{a} + \vec{b}}{|\vec{a} + \vec{b}|} = \frac{\vec{a} + \vec{b}}{2 \cos \frac{\theta}{2}}$$

$$\begin{aligned} 8. \quad \vec{b} - 2\vec{c} &= \lambda\vec{a} \\ \Rightarrow \vec{b} &= 2\vec{c} + \lambda\vec{a} \\ \Rightarrow |\vec{b}|^2 &= |2\vec{c} + \lambda\vec{a}|^2 \\ \Rightarrow 16 &= 4|\vec{c}|^2 + \lambda^2|\vec{a}|^2 + 4\lambda\vec{a} \cdot \vec{c} \\ \Rightarrow 16 &= 4 + \lambda^2 + 4\lambda \frac{1}{4} \\ \Rightarrow \lambda^2 + \lambda - 12 &= 0 \\ \Rightarrow \lambda &= 3, -4 \end{aligned}$$

$$9. \quad \text{Given } \vec{v} \cdot \vec{u} = \vec{w} \cdot \vec{u} \\ \text{and } \vec{v} \perp \vec{w} \Rightarrow \vec{v} \cdot \vec{w} = 0$$

$$\begin{aligned} \text{Now, } |\vec{u} - \vec{v} + \vec{w}|^2 &= |\vec{u}|^2 + |\vec{v}|^2 + |\vec{w}|^2 - 2\vec{u} \cdot \vec{v} - 2\vec{w} \cdot \vec{v} + 2\vec{u} \cdot \vec{w} \\ &= 1 + 4 + 9 \end{aligned}$$

$$\text{So } |\vec{u} - \vec{v} + \vec{w}| = \sqrt{14}$$

$$\begin{aligned} 10. \quad \because \vec{u} + \vec{v} + \vec{w} &= \vec{0} \\ \therefore |\vec{u} + \vec{v} + \vec{w}|^2 &= 0 \\ 9 + 16 + 25 + 2(\vec{u} \cdot \vec{v} + \vec{v} \cdot \vec{w} + \vec{w} \cdot \vec{u}) &= 0 \\ \Rightarrow \vec{u} \cdot \vec{v} + \vec{v} \cdot \vec{w} + \vec{w} \cdot \vec{u} &= -25 \end{aligned}$$

$$\begin{aligned} 11. \quad |\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2 &= 6 - 6(\cos A + \cos B + \cos C) \\ &\leq 6 - 2\left(-\frac{1}{2} - \frac{1}{2} - \frac{1}{2}\right) \end{aligned}$$

$$12. \quad \text{Given that } \vec{a} \text{ and } \vec{b} \text{ are two unit vectors} \\ |\vec{a}| = 1 \text{ and } |\vec{b}| = 1$$

$$\text{Also given that } (\vec{a} + 2\vec{b}) \perp (5\vec{a} - \vec{b})$$

$$\Rightarrow (\vec{a} + 2\vec{b}) \cdot (5\vec{a} - \vec{b}) = 0$$

$$\Rightarrow 6|\vec{a}||\vec{b}|\cos\theta = 3$$

$$[\text{where } \theta \text{ is the } \angle \text{ between } \vec{a} \text{ and } \vec{b}]$$

$$\Rightarrow \cos\theta = \frac{1}{2}$$

$$\Rightarrow \theta = 60^\circ$$

$$13. \quad \text{Vectors } \vec{a}, \vec{b}, \vec{c} \text{ and } \vec{d} \text{ are coplanar. Therefore,} \\ \sin\alpha + 2\sin 2\beta + 3\sin 3\gamma = 1.$$

$$\text{Now } |\sin\alpha + 2\sin 2\beta + 3\sin 3\gamma|$$

$$\leq \sqrt{1+4+9} \times \sqrt{\sin^2\alpha + \sin^2 2\beta + \sin^2 3\gamma}$$

$$\Rightarrow \sin^2\alpha + \sin^2 2\beta + \sin^2 3\gamma \geq \frac{1}{14}$$

$$14. \quad \text{The given relation can be rewritten as the vector expression}$$

$$(\sqrt{a^2 - 4}\hat{i} + a\hat{j} + \sqrt{a^2 + 4}\hat{k})$$

$$\cdot (\tan A\hat{i} + \tan B\hat{j} + \tan C\hat{k}) = 6a$$

$$\Rightarrow \sqrt{a^2 - 4 + a^2 + a^2 + 4}$$

$$\sqrt{\tan^2 A + \tan^2 B + \tan^2 C} \times (\cos\theta) = 6a$$

$$[\because \vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta]$$

$$\Rightarrow \sqrt{3}a \sqrt{\tan^2 A + \tan^2 B + \tan^2 C} \times (\cos \theta) = 6a$$

$$\Rightarrow \tan^2 A + \tan^2 B + \tan^2 C = 12 \sec^2 \theta \geq 12$$

$$[\because \sec^2 \theta \geq 1]$$

the least value of $\tan^2 A + \tan^2 B + \tan^2 C$ is 12

15. Let $\vec{r} = x_1 \hat{a} + x_2 \hat{b} + x_3 (\hat{a} \times \hat{b})$

$$\Rightarrow \vec{r} \cdot \hat{a} = x_1 + x_2 \hat{a} \cdot \hat{b} + x_3 \hat{a} \cdot (\hat{a} \times \hat{b}) = x_1$$

$$\text{Also, } \vec{r} \cdot \hat{b} = x_1 \hat{a} \cdot \hat{b} + x_2 + x_3 \hat{b} \cdot (\hat{a} \times \hat{b}) = x_2$$

$$\text{and } \vec{r} \cdot (\hat{a} \times \hat{b}) = x_1 \hat{a} \cdot (\hat{a} \times \hat{b}) + x_2 \hat{b} \cdot (\hat{a} \times \hat{b})$$

$$+ x_3 (\hat{a} \times \hat{b}) \cdot (\hat{a} \times \hat{b}) = x_3$$

$$\Rightarrow \vec{r} = (\vec{r} \cdot \hat{a}) \hat{a} + (\vec{r} \cdot \hat{b}) \hat{b} + (\vec{r} \cdot (\hat{a} \times \hat{b})) (\hat{a} \times \hat{b})$$

16. Let the new position of $\vec{OA}$ be $\vec{r}$. Since $\vec{r}$ is coplanar with $\vec{OA}$ and $\vec{i}$, Let

$$\vec{r} = x\vec{OA} + y\vec{i}$$

$$= (x+y)\vec{i} + 2x\vec{j} + 2x\vec{k}$$

Since $\vec{r}$ is perpendicular to $\vec{OA}$, we have

$$\vec{r} \cdot \vec{OA} = 0$$

$$\Rightarrow (x+y) + 4x + 4x = 0 \Rightarrow y = -9x$$

$$|\vec{r}| = 3 \Rightarrow (x+y)^2 + 4x^2 + 4x^2 = 9$$

$$\Rightarrow 64x^2 + 8x^2 = 9$$

$$\Rightarrow x = \pm \frac{1}{2\sqrt{2}} \text{ so that } y = \mp \frac{9}{2\sqrt{2}}$$

$$\text{Now } \vec{r} = \frac{1}{\sqrt{2}} (4\vec{i} - \vec{j} - \vec{k}) \text{ or}$$

$$\vec{r} = \frac{1}{\sqrt{2}} (-4\vec{i} + \vec{j} + \vec{k})$$

17. $M = \sqrt{(\cos^2 t + \sin^2 t + 2\hat{a} \cdot \hat{b} \sin t \cos t)}$

$$= \sqrt{1 + \hat{a} \cdot \hat{b}}$$

$$\sin 2t \text{ is max at } t = \frac{\pi}{4}$$

$$\hat{u} = \frac{\hat{a} \frac{1}{\sqrt{2}} + \hat{b} \frac{1}{\sqrt{2}}}{\left| \frac{\hat{a}}{\sqrt{2}} + \frac{\hat{b}}{\sqrt{2}} \right|} = \frac{\hat{a} + \hat{b}}{|\hat{a} + \hat{b}|}$$

18. $\vec{a} \times (\hat{i} + 2\hat{j} + \hat{k}) = \hat{i} - \hat{k} = [\hat{j} \times (\hat{i} + 2\hat{j} + \hat{k})]$

$$\Rightarrow (\vec{a} - \hat{j}) \times (\hat{i} + 2\hat{j} + \hat{k}) = \vec{0}$$

$$\Rightarrow \vec{a} - \hat{j} = \lambda (\hat{i} + 2\hat{j} + \hat{k})$$

$$\Rightarrow \vec{a} = \lambda \hat{i} + (2\lambda + 1)\hat{j} + \lambda \hat{k}, \lambda \in R$$

19. A vector perpendicular to the plane of O, P and Q is $\vec{OP} \times \vec{OQ}$

Now,

$$\vec{OP} \times \vec{OQ} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & 1 & \lambda \\ 2 & -1 & \lambda \end{vmatrix} = 2\lambda \hat{i} - 2\lambda \hat{j} - 6\hat{k}$$

Since, $\hat{i} - \hat{j} + 6\hat{k}$ is parallel to $2\lambda \hat{i} - 2\lambda \hat{j} - 6\hat{k}$.

$$\text{Hence, } \frac{1}{2\lambda} = \frac{-1}{-2\lambda} = \frac{6}{-6} = \lambda = -\frac{1}{2}$$

20. $|\vec{a} \times \vec{b} - \vec{a} \times \vec{c}| = |\vec{a} \times (\vec{b} - \vec{c})|^2$

$$= |\vec{a}|^2 |\vec{b} - \vec{c}|^2 - (\vec{a} \cdot (\vec{b} - \vec{c}))^2 = |\vec{b} - \vec{c}|^2$$

$$= |\vec{b}|^2 + |\vec{c}|^2 - 2|\vec{b}||\vec{c}|\cos \frac{\pi}{3} = 1$$

21. $1 + 9(\vec{a} \cdot \vec{b})^2 - 6(\vec{a} \cdot \vec{b}) + 4|\vec{a}|^2$

$$+ |\vec{b}|^2 + 9|\vec{a} \times \vec{b}|^2 + 4\vec{a} \cdot \vec{b} = 47$$

$$\Rightarrow 1 + 4 + 4 + 36 - 4\cos \theta = 47$$

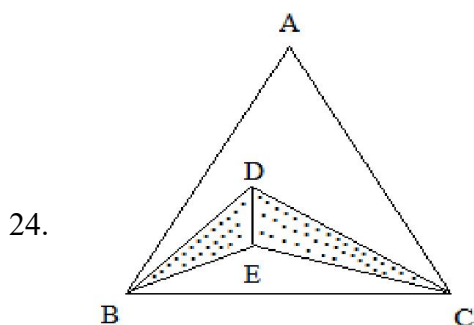
$$\Rightarrow \cos \theta = -\frac{1}{2}$$

$$\Rightarrow \text{Angle between } \vec{a} \text{ and } \vec{b} \text{ is } \frac{2\pi}{3}$$

22. $|2(\vec{a} \times \vec{b})| + 3|\vec{a} \cdot \vec{b}| \leq \sqrt{|\vec{a} \times \vec{b}|^2 + |\vec{a} \cdot \vec{b}|^2} \times \sqrt{4 + 9}$

$$\leq 6\sqrt{13}$$

$$\begin{aligned} 23. \quad \Delta &= \frac{1}{2} \left| (\hat{j} + \lambda \hat{k}) \times (\hat{i} + \lambda \hat{k}) \right| = \frac{1}{2} \left| -\hat{k} + \lambda \hat{i} + \lambda \hat{j} \right| \\ &= \frac{1}{2} \sqrt{2\lambda^2 + 1} \\ \Rightarrow \frac{9}{4} &\leq \frac{1}{4} (2\lambda^2 + 1) \leq \frac{33}{4} \\ \Rightarrow 4 &\leq \lambda^2 \leq 16 \quad \Rightarrow 2 \leq |\lambda| \leq 4 \end{aligned}$$



$$\begin{aligned} \text{We have } \overrightarrow{BC} &= \overrightarrow{BA} + \overrightarrow{AC} \\ &= (-3\hat{i} + \hat{j}) + (2\hat{i} + 3\hat{j}) = -\hat{i} + 4\hat{j} \\ \text{Vector area of the shaded region is given by} \\ \frac{1}{2} \overrightarrow{ED} \times \overrightarrow{EB} + \frac{1}{2} \overrightarrow{EC} \times \overrightarrow{ED} \\ &= \frac{1}{2} [\overrightarrow{ED} \times (\overrightarrow{EB} - \overrightarrow{EC})] \\ &= \frac{1}{2} (\overrightarrow{ED} \times \overrightarrow{CB}) = \frac{1}{2} (16\hat{k} - 2\hat{k}) = 7\hat{k} \end{aligned}$$

$$\text{Area} = |\text{Vector Area}| = 7$$

$$\begin{aligned} 25. \quad (\text{C}) \quad (\vec{a} \times \vec{b}) \times \vec{a} &= (\vec{a} \cdot \vec{a}) \vec{b} - (\vec{a} \cdot \vec{b}) \vec{a} \\ (\hat{j} - \hat{k}) \times (\hat{i} + \hat{j} + \hat{k}) &= (\sqrt{3})^2 (\vec{b}) - (\hat{i} + \hat{j} + \hat{k}) \\ \Rightarrow 3\hat{b} &= 3\hat{i} \Rightarrow \hat{b} = \hat{i} \end{aligned}$$

$$26. \quad \text{Since } \vec{a}, \vec{b}, \vec{c} \text{ are unit vectors and } \vec{a} + \vec{b} + \vec{c} = \vec{0},$$

$$\vec{a}, \vec{b}, \vec{c}$$

represent an equilateral triangle

$$\therefore \vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a} \neq \vec{0}$$

$$27. \quad \text{Let } \vec{c} = (2\hat{i} + 3\hat{j} + 4\hat{k})$$

$$\vec{a} \times \vec{c} = \vec{c} \times \vec{b}$$

$$\Rightarrow (\vec{a} + \vec{b}) \times \vec{c} = \vec{0}$$

$$\Rightarrow (\vec{a} + \vec{b}) \parallel \vec{c}$$

$$\text{Let } (\vec{a} + \vec{b}) = \lambda \vec{c}$$

$$\Rightarrow |\vec{a} + \vec{b}| = |\lambda| |\vec{c}|$$

$$\Rightarrow \sqrt{29} = |\lambda| \cdot \sqrt{29}$$

$$\Rightarrow \lambda = \pm 1$$

$$\therefore \vec{a} + \vec{b} = \pm (2\hat{i} + 3\hat{j} + 4\hat{k})$$

Now

$$(\vec{a} + \vec{b}) \cdot (7\hat{i} + 2\hat{j} + 3\hat{k}) = \pm (-14 + 6 + 12) = \pm 4$$

$$28. \quad (\text{B}) \text{ Given } \vec{a} + \vec{b} + \vec{c} = \vec{0} \text{ (by triangle law)}$$

Taking cross product by $\vec{a}$

$$\vec{a} \times (\vec{a} + \vec{b} + \vec{c}) = \vec{a} \times \vec{0} = \vec{0}$$

$$\Rightarrow \vec{a} \times \vec{a} + \vec{a} \times \vec{b} + \vec{a} \times \vec{c} = \vec{0}$$

$$\Rightarrow \vec{a} \times \vec{b} = \vec{c} \times \vec{a} \quad [\because \vec{a} \times \vec{a} = \vec{0}]$$

$$\text{Similarly, } \vec{a} \times \vec{b} = \vec{b} \times \vec{c}$$

$$\text{Therefore } \vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a}$$

$$29. \quad |\overrightarrow{BA}|^2 + t^2 |\overrightarrow{BC}|^2 - 2\overrightarrow{BA} \cdot \overrightarrow{BC} t - |\overrightarrow{AC}|^2 \geq 0$$

$$\Rightarrow |\overrightarrow{BC}|^2 t^2 - (2\overrightarrow{BA} \cdot \overrightarrow{BC}) t + (|\overrightarrow{BA}|^2 - |\overrightarrow{AC}|^2) \geq 0$$

$$\text{since, } t \in \mathbb{R} \Rightarrow \Delta \leq 0$$

$$\Rightarrow (\overrightarrow{BA} \cdot \overrightarrow{BC})^2 - |\overrightarrow{BC}|^2 |\overrightarrow{BA}|^2 + 4|\overrightarrow{BC}|^2 |\overrightarrow{AC}|^2 \leq 0 \quad \text{---(1)}$$

$$\Rightarrow (\overrightarrow{BA} \times \overrightarrow{BC})^2 + 4|\overrightarrow{BC}|^2 |\overrightarrow{AC}|^2 \leq 0$$

$$\Rightarrow -|(\overrightarrow{BC} + \overrightarrow{CA}) \times \overrightarrow{BC}|^2 + 4|\overrightarrow{BC}|^2 |\overrightarrow{AC}|^2 \leq 0$$

$$\Rightarrow |\overrightarrow{AC} \times \overrightarrow{BC}|^2 \geq |\overrightarrow{AC}|^2 |\overrightarrow{BC}|^2$$

$$\Rightarrow \sin^2 C \geq 1 \Rightarrow \sin C = 1 \Rightarrow \angle C = 90^\circ$$

$$30. \quad \vec{c} = x\vec{a} + y\vec{b} + z(\vec{a} \times \vec{b})$$

$\vec{a} \cdot \vec{c} = \vec{b} \cdot \vec{c} = \cos \theta$ [as $\vec{a}, \vec{b}$ and $\vec{c}$ are unit vectors]

$$\Rightarrow \vec{a} \cdot \vec{c} = x|\vec{a}|^2 \Rightarrow x = \cos \theta$$

$$\text{Also, } \vec{b} \cdot \vec{c} = x|\vec{b}|^2 = \cos \theta$$

$$\Rightarrow \vec{c} = \cos \theta (\vec{a} + \vec{b}) + y(\vec{a} \times \vec{b})$$

$$\text{Now, } |\vec{c}|^2 = 1 = (\cos \theta (\vec{a} + \vec{b}) + y(\vec{a} \times \vec{b})) \cdot (\cos \theta (\vec{a} + \vec{b}) + y(\vec{a} \times \vec{b}))$$

$$\Rightarrow 1 = \cos^2 \theta (\vec{a} + \vec{b}) \cdot (\vec{a} + \vec{b}) + y^2 (\vec{a} \times \vec{b}) \cdot (\vec{a} \times \vec{b})$$

$$\Rightarrow 1 = \cos^2 \theta (|\vec{a}|^2 + |\vec{b}|^2) + y^2 |\vec{a}|^2 |\vec{b}|^2$$

$$\Rightarrow 1 = 2 \cos^2 \theta + y^2$$

$$\Rightarrow \cos^2 \theta = \frac{1}{2} - \frac{y^2}{2} \Rightarrow 0 \leq \cos^2 \theta \leq \frac{1}{2}$$

$$\Rightarrow -\frac{1}{\sqrt{2}} \leq \cos \theta \leq \frac{1}{\sqrt{2}}$$

$$\Rightarrow \theta \in \left[\frac{\pi}{4}, \frac{3\pi}{4} \right]$$

$$\begin{aligned} 31. & ((\vec{a} \times \vec{b}) + (\vec{a} \times \vec{c})) \times (\vec{b} \times \vec{c}) \\ &= (\vec{a} \times \vec{b}) \times (\vec{b} \times \vec{c}) + (\vec{a} \times \vec{c}) \times (\vec{b} \times \vec{c}) \\ &= ((\vec{a} \times \vec{b}) \cdot \vec{c}) \vec{b} - ((\vec{a} \times \vec{b}) \cdot \vec{b}) \vec{c} + ((\vec{a} \times \vec{c}) \cdot \vec{c}) \vec{b} \\ &\quad - ((\vec{a} \times \vec{c}) \cdot \vec{b}) \vec{c} = [\vec{a} \vec{b} \vec{c}] (\vec{b} + \vec{c}) \\ &\Rightarrow ((\vec{a} \times \vec{b}) + (\vec{a} \times \vec{c})) \times (\vec{b} \times \vec{c}) \cdot (\vec{b} - \vec{c}) \\ &= [\vec{a} \vec{b} \vec{c}] (\vec{b} + \vec{c}) \cdot (\vec{b} - \vec{c}) \\ &= [\vec{a} \vec{b} \vec{c}] (|\vec{b}|^2 - |\vec{c}|^2) = 0 \end{aligned}$$

32. Let $\vec{a} \times \vec{b} = x\hat{i} + y\hat{j} + z\hat{k}$. Therefore,

$$[\vec{a} \vec{b} \hat{i}] = (\vec{a} \times \vec{b}) \cdot \hat{i} = x$$

$$[\vec{a} \vec{b} \hat{j}] = (\vec{a} \times \vec{b}) \cdot \hat{j} = y$$

$$[\vec{a} \vec{b} \hat{k}] = (\vec{a} \times \vec{b}) \cdot \hat{k} = z$$

$$\begin{aligned} \text{Hence, } [\vec{a} \vec{b} \hat{i}] \hat{i} + [\vec{a} \vec{b} \hat{j}] \hat{j} + [\vec{a} \vec{b} \hat{k}] \hat{k} \\ = x\hat{i} + y\hat{j} + z\hat{k} \\ = \vec{a} \times \vec{b} \end{aligned}$$

$$\begin{aligned} 33. & \vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c}) \vec{b} - (\vec{a} \cdot \vec{b}) \vec{c} \\ &= 5(\hat{i} + 2\hat{j} + 2\hat{k}) - 6(\hat{i} + \hat{j} + 2\hat{k}) \\ &\Rightarrow (1 + \alpha)\hat{i} + \beta(1 + \alpha)\hat{j} + \gamma(1 + \alpha)(1 + \beta)\hat{k} \\ &= -\hat{i} + 4\hat{j} - 2\hat{k} \\ &\Rightarrow 1 + \alpha = -1, \beta = -4 \text{ and } \gamma(-1)(-3) = -2 \\ &\Rightarrow \gamma = -\frac{2}{3} \end{aligned}$$

$$34. \begin{vmatrix} a & a & c \\ 1 & 0 & 1 \\ c & c & b \end{vmatrix} = 0 \Rightarrow \begin{vmatrix} a & a & c-a \\ 1 & 0 & 0 \\ c & c & b-c \end{vmatrix} = 0$$

$$C_3 \rightarrow C_3 - C_1$$

Expanding along R_2 we get

$$\Rightarrow c^2 - ac - ab + ac = 0 \Rightarrow c^2 = ab$$

$\Rightarrow a, c, b$ are in G.P

$\therefore c$ is the G.M of a and b

$$35. (A) \text{ Let } \vec{d} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$\text{Where } x^2 + y^2 + z^2 = 1 \quad \dots\dots\dots(i)$$

($\vec{d}$ being unit vector)

$$\therefore \vec{a} \cdot \vec{d} = 0$$

$$\Rightarrow x - y = 0 \Rightarrow x = y \quad \dots\dots\dots(ii)$$

$$\Rightarrow \begin{vmatrix} 0 & 1 & -1 \\ -1 & 0 & 1 \\ x & y & z \end{vmatrix}$$

$$\Rightarrow x + y + z = 0$$

$$\Rightarrow 2x + z = 0 \quad \dots\dots\dots(\text{using (ii)})$$

$$\Rightarrow z = -2x \quad \dots\dots\dots(iii)$$

From (i), (ii), (iii)

$$x^2 + x^2 + 4x^2 = 1$$

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$$x = \pm \frac{1}{\sqrt{6}}$$

$$36. [\vec{a} \vec{b} \vec{c}] = 0 \text{ \& } 1 + \alpha^2 + \beta^2 = 3$$

$$37. [\vec{u}, \vec{v}, \vec{w}] = -[\vec{v} \vec{u} \vec{w}] = [\vec{v} \vec{w} \vec{u}]$$

$$38. |(\vec{a} \times \vec{b}) \times \vec{c}| = |\vec{a} \times \vec{b}| |\vec{c}| \sin 30^\circ$$

$$= \frac{1}{2} |\vec{a} \times \vec{b}| |\vec{c}| \quad \dots (i)$$

$$\text{We have } \vec{a} = 2\hat{i} + \hat{j} - 2\hat{k} \text{ and } \vec{b} = \hat{i} + \hat{j}$$

$$\Rightarrow \vec{a} \times \vec{b} = 2\hat{i} - 2\hat{j} + \hat{k}$$

$$\Rightarrow |\vec{a} \times \vec{b}| = \sqrt{9} = 3$$

$$\text{Also given } |\vec{c} - \vec{a}| = 2\sqrt{2}$$

$$\Rightarrow |\vec{c} - \vec{a}|^2 = 8$$

Substituting values of $|\vec{a} \times \vec{b}|$ and $|\vec{c}|$ in (i) we get

$$|(\vec{a} \times \vec{b}) \times \vec{c}| = \frac{1}{2} \times 3 \times 1 = \frac{3}{2}$$

$$39. (a) \text{ As } \vec{c} \text{ is coplanar with } \vec{a} \text{ and } \vec{b} \text{ we take}$$

$$\vec{c} = a\vec{a} + b\vec{b}$$

.....(i)

Where a, b are scalars

$$\text{As } \vec{c} \text{ is perpendicular to } \vec{a}, \vec{c} \cdot \vec{a} = 0$$

$$\therefore \text{ From (i) we get, } 0 = a(\vec{a} \cdot \vec{a}) + b(\vec{b} \cdot \vec{a})$$

$$\Rightarrow 0 = a(6) + b(2 + 2 - 1)$$

$$\Rightarrow b = -2a$$

Thus,

$$\vec{c} = a(\vec{a} - 2\vec{b}) = a(-3\hat{j} + 3\hat{k}) = 3a(-\hat{j} + \hat{k})$$

$$\Rightarrow |\vec{c}|^2 = 9a^2(1 + 1) = 18a^2 \Rightarrow 1 = 18a^2$$

$$\Rightarrow a = \pm \frac{1}{3\sqrt{2}}$$

$$\therefore \vec{c} = \pm \frac{1}{\sqrt{2}}(-\hat{j} + \hat{k})$$

$$40. \vec{a}, \vec{b}, \vec{c} \text{ are unit coplanar vectors,}$$

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$2\vec{a} - \vec{b}, 2\vec{b} - \vec{c}$ and $2\vec{c} - \vec{a}$ are also coplanar vectors, being linear combination of $\vec{a}, \vec{b}$ and $\vec{c}$

$$\text{Thus } [2\vec{a} - \vec{b} \ 2\vec{b} - \vec{c} \ 2\vec{c} - \vec{a}] = 0$$

$$41. (C) \vec{a} = \hat{i} - \hat{k}, \vec{b} = x\hat{i} + \hat{j} + (1-x)\hat{k}$$

$$\vec{c} = y\hat{i} + x\hat{j} + (1+x-y)\hat{k}$$

$$[\vec{a} \ \vec{b} \ \vec{c}] = \begin{vmatrix} 1 & 0 & -1 \\ x & 1 & 1-x \\ y & x & 1+x-y \end{vmatrix}$$

$$= 1(1+x-y-x+x^2) - 1(x^2-y) = 1$$

$\therefore$ depends neither on x nor on y.

$$42. \text{ Given that } \vec{v} = 2\hat{i} + \hat{j} - \hat{k} \text{ and } \vec{w} = \hat{i} + 3\hat{k} \text{ and } \vec{u} \text{ is a unit vector}$$

$$\therefore |\vec{u}| = 1$$

$$\text{Now } [\vec{u} \vec{v} \vec{w}] = \vec{u} \cdot (\vec{v} \times \vec{w})$$

$$\vec{u} \cdot (2\hat{i} + \hat{j} - \hat{k}) \times (\hat{i} + 3\hat{k})$$

$$= \vec{u} \cdot (3\hat{i} - 7\hat{j} - \hat{k}) = \sqrt{3^2 + 7^2 + 1^2} \cos \theta$$

which is max. when $\cos \theta = 1$

$$\therefore \text{ max. value of } [\vec{u} \vec{v} \vec{w}] = \sqrt{59}$$

$$43. \text{ Vol. of parallelopiped formed by}$$

$$\vec{u} = \hat{i} + a\hat{j} + \hat{k}, \vec{v} = \hat{j} + a\hat{k}, \vec{w} = a\hat{i} + \hat{k} \text{ is}$$

$$V = [\vec{u} \vec{v} \vec{w}] = \begin{vmatrix} 1 & a & 1 \\ 0 & 1 & a \\ a & 0 & 1 \end{vmatrix}$$

$$= 1(1-0) - a(0-a^2) + 1(0-a)$$

$$= 1 + a^3 - a$$

$$\text{For } V \text{ to be min } \frac{dV}{da} = 0 \Rightarrow 3a^2 - 1 = 0$$

$$\Rightarrow a = \pm \frac{1}{\sqrt{3}}$$

$$\text{But } a > 0 \Rightarrow a = \frac{1}{\sqrt{3}}$$

44. (C) Any vector coplanar to $\vec{a}$ and $\vec{b}$ can be written as $\vec{r} = \vec{a} + \lambda \vec{b}$

$$\vec{r} = (1+2\lambda)\hat{i} + (-1+\lambda)\hat{j} + (1+\lambda)\hat{k}$$

since $\vec{r}$ is orthogonal to $5\hat{i} + 2\hat{j} + 6\hat{k}$

$$\Rightarrow 5(1+2\lambda) + 2(-1+\lambda) + 6(1+\lambda) = 0$$

$$\Rightarrow 9 + 18\lambda = 0 \Rightarrow \lambda = -\frac{1}{2} \quad \therefore \vec{r} \text{ is } 3\hat{j} - \hat{k}$$

$\therefore \vec{r}$ is a unit vector

$$\therefore \hat{r} = \frac{3\hat{j} - \hat{k}}{\sqrt{10}}$$

45. A vector in the plane of $\vec{a}$ and $\vec{b}$ is

$$\vec{u} = \vec{a} + \lambda \vec{b} = (1+\lambda)\hat{i} + (2-\lambda)\hat{j} + (1+\lambda)\hat{k}$$

$$\text{Projection of } \vec{u} \text{ on } \vec{c} = \frac{1}{\sqrt{3}} \Rightarrow \frac{\vec{u} \cdot \vec{c}}{|\vec{c}|} = \frac{1}{\sqrt{3}}$$

$$\Rightarrow \vec{u} \cdot \vec{c} = 1$$

$$\Rightarrow |1+\lambda + 2-\lambda - 1-\lambda| = 1$$

$$\Rightarrow |2-\lambda| = 1 \Rightarrow \lambda = 1 \text{ or } 3$$

$$\Rightarrow \vec{u} = 2\hat{i} + \hat{j} + 2\hat{k} \text{ or } 4\hat{i} - \hat{j} + 4\hat{k}$$

$$46. \begin{vmatrix} -\lambda^2 & 1 & 1 \\ 1 & -\lambda^2 & 1 \\ 1 & 1 & -\lambda^2 \end{vmatrix} = 0$$

$$\Rightarrow \lambda^6 - 3\lambda^2 - 2 = 0$$

$$\Rightarrow (1+\lambda^2)^2 (\lambda^2 - 2) = 0 \Rightarrow \lambda = \pm\sqrt{2}$$

47. $\vec{a} \times (\vec{a} \times \vec{b}) = \vec{c} \Rightarrow |\vec{a}| |\vec{a} \times \vec{b}| = |\vec{c}|$

$$\left[\because \vec{a} \perp (\vec{a} \times \vec{b}) \right]$$

$$1(1 \times 5) \sin \theta = 3 \Rightarrow \sin \theta = \frac{3}{5} \Rightarrow \tan \theta = \frac{3}{4}$$

48. $\vec{a} \times \vec{b} = \vec{c}$

$$\Rightarrow \vec{a} \times (\vec{a} \times \vec{b}) = \vec{a} \times \vec{c}$$

$$\Rightarrow (\vec{a} \cdot \vec{b}) \vec{a} - |\vec{a}|^2 \vec{b} = \vec{a} \times \vec{c}$$

$$\Rightarrow \vec{b} = \frac{\beta \vec{a} - \vec{a} \times \vec{c}}{|\vec{a}|^2} \left[\because \vec{a} \cdot \vec{b} = \beta \right]$$

49. $\vec{a} = \vec{b} + \vec{c}, \vec{b} \times \vec{d} = \vec{0}, \vec{c} \cdot \vec{d} = 0$

$$\Rightarrow \vec{a} \times \vec{d} = \vec{b} \times \vec{d} + \vec{c} \times \vec{d} = \vec{c} \times \vec{d}$$

$$\Rightarrow \vec{d} \times (\vec{a} \times \vec{d}) = \vec{d} \times (\vec{c} \times \vec{d})$$

$$= (\vec{d} \cdot \vec{d}) \vec{c} - (\vec{c} \cdot \vec{d}) \vec{d} = |\vec{d}|^2 \vec{c}$$

$$\Rightarrow \vec{c} = \frac{\vec{d} \times (\vec{a} \times \vec{d})}{|\vec{d}|^2}$$

50. We have $\vec{a} \times (\vec{b} \times \vec{c}) = \frac{\vec{b} + \vec{c}}{\sqrt{2}}$

$$\Rightarrow (\vec{a} \cdot \vec{c}) \vec{b} - (\vec{a} \cdot \vec{b}) \vec{c} = \frac{\vec{b}}{\sqrt{2}} + \frac{\vec{c}}{\sqrt{2}}$$

$$\Rightarrow \left(\vec{a} \cdot \vec{c} - \frac{1}{\sqrt{2}} \right) \vec{b} - \left(\vec{a} \cdot \vec{b} + \frac{1}{\sqrt{2}} \right) \vec{c} = 0$$

$$\Rightarrow \text{then } \vec{a} \cdot \vec{c} - \frac{1}{\sqrt{2}} = 0 \text{ and } \vec{a} \cdot \vec{b} + \frac{1}{\sqrt{2}} = 0$$

$$\therefore \vec{a} \cdot \vec{b} = -\frac{1}{\sqrt{2}} \Rightarrow (\vec{a}, \vec{b}) = \frac{3\pi}{4}$$

51. $(\vec{a} + \vec{b} + \vec{c}) \cdot [(\vec{a} + \vec{b}) \times (\vec{a} + \vec{c})]$

$$= (\vec{a} + \vec{b} + \vec{c}) \cdot [\vec{a} \times \vec{a} + \vec{a} \times \vec{c} + \vec{b} \times \vec{a} + \vec{b} \times \vec{c}]$$

$$= \{ \vec{a} \cdot (\vec{a} \times \vec{c}) + \vec{a} \cdot (\vec{b} \times \vec{a}) + \vec{a} \cdot (\vec{b} \times \vec{c}) \}$$

$$+ \{ \vec{b} \cdot (\vec{a} \times \vec{c}) + \vec{b} \cdot (\vec{b} \times \vec{a}) + \vec{b} \cdot (\vec{b} \times \vec{c}) \}$$

$$+ \{ \vec{c} \cdot (\vec{a} \times \vec{c}) + \vec{c} \cdot (\vec{b} \times \vec{a}) + \vec{c} \cdot (\vec{b} \times \vec{c}) \}$$

$$= [\vec{a} \vec{b} \vec{c}] - [\vec{a} \vec{b} \vec{c}] - [\vec{a} \vec{b} \vec{c}] = -[\vec{a} \vec{b} \vec{c}]$$

52. (B) As p, q and r are three mutually perpendicular vectors of same magnitude, so let us

$$\text{consider } \vec{p} = a\hat{i}, \vec{q} = a\hat{j}, \vec{r} = a\hat{k}$$

$$\text{Also let } \vec{x} = x_1\hat{i} + y_1\hat{j} + z_1\hat{k}$$

Given that $\vec{x}$ satisfies that equation

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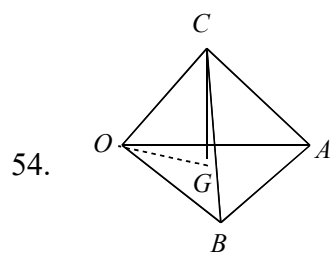
$$\begin{aligned}
 & \vec{p} \times [(\vec{x} - \vec{q}) \times \vec{p}] + \vec{q} \times [(\vec{x} - \vec{r}) \times \vec{q}] \\
 & + \vec{r} \times [(\vec{x} - \vec{p}) \times \vec{r}] = 0 \quad \dots\dots(i) \\
 & \Rightarrow 3\vec{a}^2\vec{x} - \vec{a}^2\vec{x} - \vec{a}^2(\vec{p} + \vec{q} + \vec{r}) = 0 \\
 & \Rightarrow \vec{x} = \frac{1}{2}(\vec{p} + \vec{q} + \vec{r})
 \end{aligned}$$

53. If θ is the angle between P_1 and P_2

$$\begin{aligned}
 & \left. \begin{aligned} N_1 &= \vec{a} \times \vec{b} \\ N_2 &= \vec{c} \times \vec{d} \end{aligned} \right\} N_1 \times N_2 = 0
 \end{aligned}$$

$$\text{then } |N_1| \times |N_2| \sin \theta = 0$$

$$\text{or } \sin \theta = 0 \Rightarrow \theta = 0$$



$$|a| = |b| = |c| = 1$$

$$\overrightarrow{OC} = \overrightarrow{OG} + \overrightarrow{GC}$$

$$\text{But } \overrightarrow{OG} = \frac{\vec{a} + \vec{b}}{3}$$

$$\text{and } \overrightarrow{GC} = |\overrightarrow{GC}| \left(\frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|} \right)$$

$$= \frac{2\sqrt{2}}{3} (\vec{a} \times \vec{b})$$

$$\therefore \frac{1}{6} [\vec{a} \vec{b} \vec{c}] = \frac{1}{3} \left[\frac{1}{2} |\vec{a} \times \vec{b}| \right] \times |\overrightarrow{GC}|$$

$$\therefore \overrightarrow{OC} = \frac{\vec{a} + \vec{b}}{3} + \frac{2\sqrt{2}}{3} (\vec{a} \times \vec{b}) \Rightarrow 3\overrightarrow{OC} = \vec{a} + \vec{b} + 2\sqrt{2} (\vec{a} \times \vec{b})$$

